The Spillover Effects of Federal Aid Programs: Evidence from Flood Insurance and Municipal Borrowing

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Abstract

This paper examines how federal programs targeting individuals can generate economic spillovers for local governments. I study the spillover effects of social welfare programs on municipal bond markets, using local variation in the rollout of a federal program providing subsidized flood insurance to households. I find that municipalities with broader program access experience smaller declines in property tax revenue following floods. I then show that areas with greater access also face lower financing costs. The effects are most pronounced in high-risk and high-income counties. Overall, these results show how federal subsidies can produce positive local externalities, enhancing fiscal stability.

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1 Introduction

Economic shocks are often local, but the programs that insure against them are often national. Floods, job losses, housing instability, and other disasters are adverse shocks to households in particular places, while many programs that absorb these losses are financed and administered federally. This separation matters because household recovery can affect the local tax base. When federally financed programs help residents absorb losses, their effects may extend beyond direct beneficiaries to the local governments whose revenues depend on those households. The direction of the spillover effect is unclear ex ante. Insurance that helps households recover after adverse income shocks may preserve property values, stabilize revenues, and lower municipalities' borrowing costs. But insurance may also change household behavior, for instance by delaying job searches or encouraging households to remain in or move to flood-prone areas, in ways that weaken local economic conditions and raise municipal borrowing costs (Moffitt 1992; Krueger and Meyer 2002; Chetty 2008). This paper examines whether the National Flood Insurance Program (NFIP), a federal program that subsidizes household flood insurance, affects the cost of borrowing for the local governments whose tax bases depend on those households and properties.

Floods are among the most destructive natural disasters in the United States, with damages frequently reaching billions of dollars. Since 2013, the number of billion-dollar flood events has nearly doubled relative to the prior decade (NOAA National Centers for Environmental Information 2024). The rising costs are reflected in the NFIP's financial health. As of February 2025, the NFIP carries $22.5 billion in outstanding debt to the Treasury, even after Congress canceled $16 billion of its debt following Hurricanes Harvey, Irma, and Maria (Horn and Webel 2025). The program is the largest flood insurance provider in the United States, accounting for 95% of flood insurance policies and providing more than $1.3 trillion in coverage across about 4.7 million policies (Kousky et al. 2018; Horn and Webel 2025).

These figures illustrate the scale of federal funds allocated to insure households against flood risk, but they do not measure whether federal insurance also affects the local governments that depend on those households and properties for revenue. If flood insurance helps homeowners repair damaged properties after floods, it may preserve property values, stabilize property tax revenues, and lower the default risk priced by municipal bond investors. If insurance instead lowers the private cost of living in flood-prone areas, it may encourage additional development, increasing local governments' future exposure to flood losses by placing more property, infrastructure, and tax revenue at risk (Kane and Puentes 2015;

Peralta and Scott 2024).[1] Federal officials recognized this concern as early as 1982, warning that subsidized flood insurance could encourage development in high flood risk areas (U.S. Government Accountability Office 1982). This paper tests whether federally subsidized flood insurance for individuals has measurable effects on local financial markets, and whether those effects reflect revenue stabilization or greater exposure to future flood losses.

A key identification challenge is that flood insurance take-up may be correlated with unobservable county characteristics. To address this concern, my first identification strategy uses the staggered rollout of flood maps across communities as a proxy for changes in flood insurance access over time. Households can purchase subsidized flood insurance only after the Federal Emergency Management Agency (FEMA) issues a flood map to their community, which is a local jurisdiction within a county. Because mapping occurs through a federal process not controlled by local governments' financing decisions, the estimates are less likely to reflect local economic changes that independently affect borrowing costs.

My second identification strategy uses variation in the timing and severity of flood events. I test whether borrowing costs rise less after floods in communities that have access to subsidized flood insurance. If insurance protects households and the local tax base, then access should dampen the financial effects of flood damage relative to otherwise similar communities without access. The identifying assumption is that, absent access to subsidized flood insurance, flood events would have affected borrowing costs similarly in communities with and without access. This assumption would be violated if FEMA mapped communities in anticipation of specific future floods or local credit deterioration. I address this concern by measuring insurance access 12 months before bond issuance. FEMA also reported that it prioritizes communities based on historical flood risk (Grimm 2020). The two identification strategies provide complementary sources of variation, one from the rollout of insurance availability and the other from the occurrence and severity of flood events.

I estimate the effect of flood insurance access on county and community borrowing costs using general obligation bonds issued between 1969 and 2019, ending before the onset of the COVID-19 pandemic. My primary outcome variable is a community's stated all-in cost of borrowing, which has a 90% correlation with initial yields when both are reported. Unlike yield-to-maturity or true interest cost, which are sparsely reported between 1969 and 1990, the stated all-in cost is reported for more than 1,600 community-month observations per year on average during that period. I measure insurance access using the date on which a community receives its first flood map. For community outcomes, access is an indicator of

[1] Development in flood-prone areas can itself raise flood exposure and the probability of damaging floods, by increasing impervious surface area and building in the floodplain and by intensifying local rainfall (Gori et al. 2019; Zhang et al. 2018; Brody et al. 2015).

whether the community has been mapped by the bond sale date. For county outcomes and aggregate specifications, I define access as the share of communities within the county that have been mapped by a given date. This measure is necessary for property tax revenue, which is observed at the county level, and useful for estimating the effect of moving from no access to full access within a county.

I first test whether access to flood insurance affects local governments' underlying revenues. The mechanism is as follows. Floods damage homes and reduce property values. Insurance payouts help homeowners repair damaged property, which should limit declines in the local property tax base. To test this channel, I estimate whether county-level insurance access mitigates the decline in property tax revenue following flood events. Across counties in the same state, I find that a one standard deviation increase in flood events lowers annual property tax revenue by 1.3% in a county with no insurance access. At the average level of county access, where about half of communities are mapped, flood insurance access offsets the estimated decline. The results are similar when flood exposure is measured using flood-related property damage rather than the number of flood events. For the median county, the implied revenue stabilization is about $200,000 per one standard deviation increase in flood events, comparable to the median county's entire annual budget for libraries or fire protection.

Stabilizing the local tax base should also be priced in municipal credit markets. If flood insurance reduces post-disaster declines in tax revenue, investors may view municipal debt as safer and require a lower rate of return. Consistent with this prediction, I find that insurance access reduces stated all-in costs by about 4 basis points within communities. County-level estimates measure the broader effect of access across communities that share the same property tax base. At the county level, full access to flood insurance reduces the county-average stated all-in cost by about 17 basis points, or 3% of the unconditional sample mean of 5.49%. This reduction is comparable to common municipal bond pricing discounts associated with bank qualification and credit enhancement. However, bank qualification reduces yields through favorable tax treatment as long as eligible issuers remain sufficiently small, while credit enhancement reduces yields through third-party repayment support that issuers typically purchase through premiums or fees. From the perspective of local governments, federally subsidized flood insurance instead produces a comparable reduction in borrowing costs that the municipality neither chooses nor pays for, and that operates outside the bond contract. The implied aggregate magnitude is economically large. Across the sample, applying the estimate for full insurance access to outstanding county general obligation debt

implies annual interest savings of about $100 to $200 million.[2] For comparison, the NFIP runs an average annual deficit of about $720 million.[3]

An alternative mechanism is that flood maps affect investors' perceptions of flood risk. But if maps mainly signal that an area faces high flood risk, they should raise perceived default risk and increase yields. This prediction is the opposite of what I find. Total annual issuance increases by about 3% after a community gains insurance access, suggesting that local governments take advantage of cheaper funding. The fact that stated all-in costs fall despite higher issuance also suggests that the decline in perceived default risk from greater insurance access more than offsets any upward pressure on borrowing costs from additional bond issuance. These findings suggest that subsidized flood insurance indirectly reduces municipalities' issuance costs even in years without floods.

A potential concern for identification is that earlier mapping may correlate with stronger local economies or better public finances. Several features of the setting mitigate this concern. The exact timing of when a community receives its map is mainly dictated by federal administrative capacity, including contractor availability, hydrological data collection, and appropriations cycles (American Institutes for Research 2005; Knowles and Kunreuther 2014; Grimm 2020; U.S. Government Accountability Office 2021). The empirical design absorbs persistent local characteristics and conditions on time-varying population, income, employment, and flood risk. As a result, the variation in mapping that I exploit is driven by these administrative delays, which are plausibly separate from local governments' financing needs. The findings are robust to dropping the earliest mapped communities and to using the Callaway and Sant'Anna (2021) adjustment for staggered treatment timing. Last, assigning random mapping dates to communities produces no effect on borrowing costs.

If flood insurance stabilizes property tax revenues, then its effect on borrowing costs should be strongest after floods, when revenue disruptions are most likely. These tests use the second strategy described above. I measure recent flood exposure using either the number of flood events or the amount of flood-related property damage in the 12 months before bond issuance. A one standard deviation increase in the number of flood events increases borrowing costs by about 2 basis points. In mapped communities, about 70% of this increase is offset

[2]I compute annual interest savings as $S = D \times \beta$, where β is the county-level full-access coefficient of 17 basis points and D is the outstanding stock of county general obligation debt. Assuming issuance is broadly constant over time, the stock equals annual issuance times the average life of the debt, $D \approx Q \times \bar{a}$, where annual issuance Q = $8.2 billion ($5.31 million average monthly issuance per community × 78,763 observations ÷ 51 years). The maturity reported in Table 1 is final maturity, with a mean of 14.94 years. Under level serial amortization the average life is about half the final maturity, so $\bar{a} \approx 7.5$ years implies $D \approx$ $62 billion and $S \approx$ $105 million per year. Treating every issue as a term bond repaid at final maturity gives the upper bound $\bar{a} = 14.94$ years, $D \approx$ $123 billion, and $S \approx$ $209 million.

[3]See Risk Rating 2.0 program brief (`https://www.fema.gov/sites/default/files/documents/fema_rr-2.0_04-2025.pdf`).

relative to unmapped communities. Unmapped communities also issue more debt after flood events than mapped communities, consistent with flood insurance reducing mapped communities' need for emergency borrowing during post-flood recovery. Lower borrowing costs and issuance amounts after floods suggest that flood insurance reduces both perceived default risk and the need to borrow during costly recovery periods.

To assess the distributional policy implications of flood insurance, I test where access produces the largest reductions in municipal borrowing costs. In 2018, FEMA reported that, even within the same flood zones, high-income households were more likely to hold NFIP coverage (FEMA 2018). My results are consistent with FEMA's report. Counties with higher income per capita experience larger reductions in borrowing costs from insurance access. Coastal counties, which face higher borrowing costs than inland counties, experience an additional 8 basis point reduction from access, on average. A one standard deviation increase in population density is associated with an additional reduction of about 7 basis points. These results suggest that flood insurance access lowers perceived default risk most in places with larger property tax bases and higher expected take-up. They also highlight a policy tradeoff. Expanding household insurance can stabilize local governments, but it may also weaken market signals of climate risk in coastal or heavily populated counties where those signals are most important.

To distinguish the property tax mechanism from alternative explanations, I examine two settings where the mechanism should not operate. First, I examine revenue bonds, which are backed by project-specific revenues rather than property taxes. If flood insurance lowers borrowing costs by stabilizing property values and property tax revenue, there should be no effect on revenue bonds. Consistent with this prediction, I find no statistically significant effect of flood insurance access on the borrowing costs of revenue bonds. Second, I test whether flood insurance access also mitigates the effect of weather events other than floods. Wildfires raise municipal borrowing costs (Berry-Stölzle and Hao 2026; Liao and Kousky 2022) but are not covered by the NFIP. I find that wildfires increase stated all-in costs, but flood insurance access does not dampen the increase.

The paper documents a financial spillover from federal household insurance subsidies to local governments. The existing literature on spillovers generally studies horizontal effects of one local government's spending or policy on neighboring governments at the same level (Case et al. 1993; Baicker 2005). The spillover documented here runs vertically. Federal subsidies lower the cost of flood insurance for households and, in turn, lower the borrowing costs of the jurisdictions in which those households reside, a benefit that extends to the local governments whose revenues depend on the recovery of those households and properties. Municipal borrowing costs provide a forward-looking measure of how investors value this

spillover for local government finances.

These findings also speak to a broader puzzle in climate finance. Municipal borrowing costs respond to climate risk, but the effects appear muted or delayed (Painter 2020; Goldsmith-Pinkham et al. 2023). The muted response may partly reflect federal policy. If federal insurance reduces the expected fiscal damage from floods, then municipal bond prices should reflect climate risk net of federal insurance rather than physical climate risk alone. Auh et al. (2022) show that federal disaster aid mitigates disaster-related losses in the secondary municipal bond market. I document an ex ante channel through which subsidized household insurance lowers local government borrowing costs before disasters occur and limits their rise when disasters occur. More broadly, the paper shows that federal aid to households can affect the price at which local governments borrow, even when no transfer is made directly to the local government. This differs from existing work on local spillovers of federal aid programs, which has focused on labor markets, housing markets, and transfers after disasters (Kline and Moretti 2014; Deryugina 2017; Begley et al. 2024; Hendren and Sprung-Keyser 2020). These benefits to local governments are not visible in NFIP claim payments or federal outlays, but they are priced in the municipal bond market and flow disproportionately to counties with higher incomes, coastal locations, and greater density. Federal household insurance can therefore both protect local governments from disaster risk and weaken the market signals that would otherwise reflect that risk.

The remainder of the paper is structured as follows. Section 2 reviews the related literature. Section 3 presents the conceptual framework. Section 4 provides institutional background on the NFIP. Section 5 describes the data and sample construction. Section 6 outlines the empirical strategy. Section 7 presents the main results. Section 8 reports falsification tests. Section 9 concludes.

2 Literature Review

This paper contributes to two strands of literature. First, it adds to research on climate risk, natural disasters, and municipal bond markets. Prior studies show that climate risk can affect local government financing conditions. Painter (2020) finds that counties more exposed to long-run climate risk face higher borrowing costs, and Yu and Zhang (2023) show that climate-related events increase municipal financing costs. Other work finds that the response of municipal bond markets to climate risk is muted or delayed, particularly in areas with high exposure (Goldsmith-Pinkham et al. 2023; Jerch et al. 2023). Auh et al. (2022) study the secondary market response to natural disasters and show that disasters reduce municipal bond returns, especially for uninsured revenue bonds, while federal disaster aid mitigates

these losses. I examine whether access to a federal household insurance program changes local government borrowing costs before losses are realized and dampens the increase in borrowing costs when floods occur. The results show that, even though the program formally targets households, it lowers local government borrowing costs in the primary bond market. This finding suggests that federal interventions can dampen the financial consequences of climate risk for local governments before disaster losses are realized.

Second, this paper contributes to research on federal aid programs and their spillovers beyond direct recipients. Prior work on public goods emphasizes that the governments receiving the benefits of a program should also face its costs, because separating benefits from financing can weaken incentives to manage local risk (Weisbrod 1964; Olson 1969; Oates 1972). Federal aid to households provides a setting in which benefits and financing are separated across levels of government. The program subsidizes insurance coverage for individuals to protect against tail risks while also benefiting the local governments where those individuals live. Deryugina (2017) shows that non-disaster federal transfers, including unemployment insurance, rise substantially after hurricanes, suggesting that the federal fiscal cost of natural disasters is understated when attention is limited to formal disaster aid programs. Begley et al. (2024) show that federal disaster lending has distributional consequences because SBA loan approvals differ across borrowers. More broadly, Kline and Moretti (2014) and Hendren and Sprung-Keyser (2020) show that government programs should be evaluated not only by their direct effects on recipients, but also by their spillovers, equilibrium responses, and implications for public budgets beyond direct program outlays.

This paper differs from other studies of public goods by showing that this vertical spillover is priced in financial markets. Expanding household access to subsidized flood insurance lowers municipal borrowing costs, reducing the cost of public investment in places exposed to flood risk. These benefits are not reflected in NFIP claim payments or federal outlays, but they appear in municipal bond prices. The bond market therefore provides a dollar value for an indirect benefit of household insurance that would otherwise be difficult to measure. It also reveals who receives that benefit. The largest gains flow to high-income, coastal, and densely populated communities, suggesting that the largest indirect benefits of federal household aid are directed toward places with larger tax bases and greater capacity to recover.

3 Conceptual Framework

This section develops a framework linking household flood insurance to municipal bond yields. It maps the empirical setting into a simple model of municipal bond pricing, formalizes

the primary mechanism through which household flood insurance can affect local borrowing costs, and identifies the comparative statics that motivate the empirical tests. The main mechanism is that flood insurance stabilizes the local property tax base by helping households repair damaged property after floods. Insurance lowers municipal yields in two ways. First, it reduces expected flood damage to the tax base. Second, it reduces how much the tax base varies after a flood. Both effects lower the yield investors require on municipal debt. Internet Appendix A presents a formal model that derives the predictions summarized here.

Consider a community with homeowners and a local government that finances itself in part through a property tax. The community has a taxable property base, a share of which is exposed to flood damage. A flood occurs with some probability, and conditional on a flood, exposed property loses a fraction of its value. Bondholders price general obligation bonds using both the expected level of future local tax revenues and the risk around those revenues, and both depend on flood damage to the exposed tax base.

Subsidized flood insurance enters through household coverage. After NFIP access, eligible households can purchase coverage, but coverage remains partial in practice because not all exposed households take up policies and because NFIP payouts do not cover the full economic loss from a flood. Insured households receive claim payments that help pay for repairs, while uninsured households bear more of the loss directly. Holding everything else fixed, more coverage means less flood damage is capitalized into taxable property values. Insurance access therefore reduces both the expected decline in the tax base after floods and the variability of the tax base from one year to the next. By reducing the tax base's exposure to flood losses, insurance access lowers the repayment risk priced by municipal bond investors and therefore lowers yields.

3.1 Timing

The timing of bond pricing does not depend on the timing of when flood damage reaches tax collections. Property tax assessments occur on fixed cycles, insurance claims take time to process, and repairs take longer still. This lag determines when property tax revenues move, but not necessarily when yields respond.

Municipal bond prices reflect expected future revenues. Investors can therefore price the revenue stabilizing effect of insurance before it appears in realized tax collections. The framework predicts that yields can respond when access changes, while property tax revenue effects appear with the delay built into assessment and collection cycles.

3.2 Heterogeneous Effects

The effect of floods on expected cash flows and volatility implies that the decline in yields after residents gain flood insurance access is not uniform. Instead, it is larger in communities where flood losses to the tax base are larger and more volatile and where coverage offsets more of those losses. The flood loss that insurance offsets scales with three primitives. First, a higher flood probability raises the expected loss that coverage can prevent. Second, a larger exposed share of the tax base raises the value at risk in a flood. Third, higher coverage among exposed households offsets a larger fraction of the loss. Both the expected offset and the reduction in loss volatility increase with each primitive, so flood insurance access has a stronger effect on borrowing costs when these primitives are higher.

The empirical analysis proxies for these three primitives with three county-level measures. Whether a county lies on the coast proxies for flood probability, because coastal counties face higher baseline flood risk than inland counties. Population density proxies for the exposed share, because densely populated counties concentrate more taxable value within a given flood-prone area. Income proxies for insurance coverage, because higher-income households are more likely to purchase and maintain flood insurance (FEMA 2018). The prediction is therefore that the reduction in borrowing costs from flood insurance access is concentrated in coastal, densely populated, and higher-income areas, where the revenue stabilizing channel should matter most.

4 Institutional Background

The U.S. Congress established the NFIP in 1968 because private flood insurance had become prohibitively expensive for most households due to the severity and unpredictability of flood-related losses (Anderson 1974). The program's goals are to provide access to flood insurance for households in at-risk areas and to reduce the nation's flood risk through floodplain management standards (Horn and Webel 2025). The NFIP was never intended as a permanent federal subsidy. Congress provided flood insurance at below-cost rates as an interim solution until state and local governments enacted land use regulations that would restrict development in risky areas and shift residents toward safer locations over time (Klein 2019), but the program now accounts for about 95% of residential flood insurance contracts in the United States (Kousky 2018). It currently insures about 4.7 million policies and provides more than $1.3 trillion in coverage (Horn and Webel 2025).

Households become eligible to purchase subsidized NFIP coverage only after FEMA issues a flood map for their community. A community is a county subdivision, such as a city, town, or village. After receiving a flood map, the community must accept the map and

adopt FEMA's floodplain management standards for residents to become eligible for NFIP coverage. Communities that reject the map leave residents ineligible for NFIP coverage and may also restrict access to federal disaster assistance after a flood (Horn and Webel 2025). Flood maps also delineate Special Flood Hazard Areas (SFHAs), defined as areas with a 1% or greater annual chance of flooding. These zones are used to determine insurance rates, insurance purchase requirements for some mortgage borrowers, and local building standards (FEMA 2018).

The mapping process is slow and capacity constrained. FEMA officials testified in 2020 that issuing a new flood map takes an average of seven years to complete (Grimm 2020). One year after the program began, only four communities had been mapped despite FEMA identifying more than 7,000 as having high flood risk (Anderson 1974). The average community waits about 18 years for its first flood map, and within counties that contain multiple participating communities, the median spread between the earliest and latest flood map issuance dates is 13 years. Thus, the pace at which individual communities are mapped is constrained by federal capacity and congressional appropriations for flood mapping (U.S. Government Accountability Office 2021; Morrissey 2006).

The NFIP provides coverage at below-market rates. As of 2023, the median NFIP premium is about 40% below the actuarially fair price, with the federal government covering the difference between that price and the subsidized cost (Federal Emergency Management Agency 2024). The program has been unable to cover its obligations from premium revenue alone since Hurricane Katrina in 2005. It currently owes $22.5 billion to the U.S. Treasury, and Congress canceled an additional $16 billion in obligations in 2017 to allow the program to pay claims from recent hurricanes (Horn and Webel 2025). Participation is generally voluntary, but homeowners in SFHAs with federally backed mortgages have been required to purchase flood insurance since 1973, and many private mortgage lenders also enforce this requirement (Sastry et al. 2024). Take-up remains incomplete and unevenly distributed. In a 2018 study matching NFIP records with Census data, FEMA reported that about 60% of mortgage borrowers in areas with high flood risk held flood insurance policies, and that policyholders in those areas typically had higher incomes than households without policies in the same areas (FEMA 2018).

5 Data and Sample Construction

I combine four data sources covering municipal bonds, flood insurance access, county outcomes, and weather events. Internet Appendix B contains additional data construction details.

Municipal Bond Issuance. Bond data come from the Securities Data Company (SDC) Platinum Global Public Finance U.S. New Issues database. I restrict the main sample to general obligation bonds issued between 1969 and 2019 with maturities greater than one year and a stated all-in cost below 50 percent, if populated. The stated all-in cost, defined by SDC as the average annual cost of borrowing including underwriting and other fees, is the primary outcome variable. It is the most consistently reported measure of borrowing costs in SDC before 1990 and is 90% correlated with initial yields when both are reported.[4] Revenue bonds are excluded from the main sample and used as a placebo test in Section 8.

Flood Insurance Access. Community-level flood insurance dates come from FEMA's Community Status Book, which lists NFIP-participating communities and the date each first received a flood map. At the community level, the treatment indicator equals one if the community's initial flood map was issued on or before the bond sale date. At the county level, I measure access as the equal-weighted share of communities within a county that have received a flood map by a given date, following Peralta and Scott (2024). Both measures reflect eligibility to purchase subsidized NFIP coverage rather than actual take-up. For robustness tests reported in the Internet Appendix, I collect OpenFEMA claims data covering 2.7 million NFIP claim payments from 1978 to 2019.

Matching Bond Issuers to NFIP Communities. I match SDC issuers to FEMA Community Identifiers (CIDs) in three stages: exact name match within state and county, fuzzy string match at an 85% similarity threshold, and a county-fallback assignment for issuers without a city-level match. The final match rate is 96.5%. Of the matched issuers, about 59% link to a specific incorporated community with its own flood map date, while about 38% are county-wide entities such as school districts and special authorities assigned the county's overall insurance access status. The main results hold when the sample is restricted to city-matched communities (Table C.1). This matching procedure creates the community-month panel that links each bond issuer to the flood insurance access status of its matched NFIP community.

County Fiscal Outcomes and Demographics. Property tax revenue comes from the U.S. Census Bureau's Annual Survey of State and Local Government Finances. Revenue figures are adjusted to 2020 dollars using the CPI. County-level population, per capita income, and employment come from the Bureau of Economic Analysis (BEA) and are available annually beginning in 1969. Population density is the county's annual population divided by its 1990 land area, and coastal status is taken from the National Oceanic and Atmospheric Administration (NOAA) coastal counties list.

Flood and Weather Events. County-level flood, wildfire, and snowstorm events come

[4]Figure C.1 plots the relationship between initial yields and the stated all-in cost over the sample period.

from the Spatial Hazard Events and Losses Database (SHELDUS), which reports event dates, affected counties, and inflation-adjusted property damage. I construct rolling 12-month measures of flood counts and log flood damage at the county level. The same construction is applied to wildfires and snowstorms for the placebo tests in Section 8. Monthly precipitation and temperature controls come from the PRISM Climate Group at Oregon State University.

Table 1 reports summary statistics for the resulting community-by-month-year sample of 78,763 observations across 9,641 unique bond-issuing communities.

[Insert Table 1 Here]

6 Empirical Strategy

I exploit two plausibly exogenous sources of variation to identify the effect of subsidized household insurance on municipal bond markets. The first uses the staggered rollout of FEMA flood maps across communities, which generates variation in the timing of households' access to subsidized flood insurance. Figure 1 illustrates this variation within a single county by showing the expansion of mapped communities over time. The second uses the timing and severity of floods themselves, which I interact with insurance access to test whether municipal bond markets price the protection that insurance provides when shocks occur.

Both strategies use flood map issuance as the source of variation. Households can only purchase subsidized flood insurance after their community receives a map, so the date of initial map issuance provides an intent-to-treat measure of access. This intent-to-treat approach avoids selection bias from individual insurance decisions, which may be correlated with income, risk preferences, or private information about flood exposure. The estimates therefore likely understate the full effect of flood insurance on municipalities' financial outcomes.

6.1 Flood Mapping

The first identification strategy uses the staggered timing of initial flood map issuances across communities. The community-level difference-in-differences design requires that, absent insurance access, borrowing costs in communities with and without access would have evolved similarly after conditioning on fixed effects and controls. The main concern is that the timing of map issuance may be correlated with unobserved community characteristics.

Several features of the setting mitigate this concern. As discussed in Section 4, FEMA's mapping process was constrained by federal administrative capacity. The timing with which individual communities received maps depended on contractor availability, hydrological data collection, and congressional appropriations for flood mapping. These constraints were substantial. One year after the program was introduced, only four communities had received

flood maps despite more than 7,000 being identified as high risk. Because the pace of mapping was set by federal capacity rather than by communities' financing decisions, the variation in when communities gained access is plausibly unrelated to the local conditions that drive borrowing costs.

The baseline specification compares borrowing costs before and after a community receives its first flood map. Community fixed effects absorb time-invariant differences across communities, including persistent differences in geography, baseline flood exposure, and local institutions. In specifications comparing cross-sectional community variation in insurance access within the same state, I use five-year mapping cohort fixed effects to absorb the systematic differences between early-mapped and late-mapped communities. Month-year fixed effects absorb aggregate movements in municipal borrowing costs, macroeconomic conditions, issuance seasonality, and aggregate weather patterns. I also control for time-varying local conditions, including the employment rate, income per capita, and population, along with a fixed measure of flood risk. These controls absorb local economic conditions that could otherwise be correlated with both the timing of mapping and borrowing costs. Conditional on these fixed effects and controls, the remaining variation in the timing of access comes from administrative constraints on FEMA's mapping process.

6.2 Flood Interactions

My second identification strategy uses the timing and severity of realized floods. If flood insurance lowers borrowing costs by stabilizing households and the local tax base, then the effect of access should be strongest when floods occur. I test this prediction by interacting insurance access with recent flood exposure, measured using either the number of floods or flood-related property damage in the prior 12 months.

For this interaction to identify how insurance access changes post-flood borrowing costs, the design requires that, after conditioning on community and month-year fixed effects, flood timing and severity are unrelated to unobserved determinants of borrowing costs that also vary with insurance access. Month-year fixed effects absorb aggregate weather and macroeconomic conditions, so the identifying variation is the idiosyncratic timing and severity of local floods. A remaining concern is that insurance access itself may change in anticipation of flood events, but this is unlikely because FEMA does not map communities in anticipation of specific future floods (Grimm 2020). I also lag insurance access by 12 months, so access status is determined before the flood exposure window used in the interaction. This timing convention reduces the concern that the estimates reflect map issuances immediately in response to a flood.

The two strategies analyze both the average effect of insurance access and the mechanism

through which access stabilizes local tax revenue. The rollout design examines whether expanding access to federal flood insurance changes municipal borrowing costs on average. The flood interaction design examines whether those effects are concentrated when floods occur.

7 Results

7.1 Flood Insurance Access and Property Tax Revenue

The first test examines whether access to flood insurance stabilizes local governments' underlying cash flows. Floods damage homes and reduce property values. Insurance payouts help homeowners repair damaged property, which should limit declines in the local property tax base.[5] Because property taxes are the primary source of repayment for general obligation bonds (Rankin, Osburn, and Rogers 1978), the effect of floods and insurance access on property tax revenue provides a direct measure of the revenue spillover that municipal bond markets may price.

I estimate this revenue stabilization with a staggered, continuous difference-in-differences specification:

$$Y_{i,t} = \alpha + \beta_1 \, \textit{Treated}_{i,t-1} \times \textit{Flood}_{i,t-1} + \beta_2 \, \textit{Treated}_{i,t-1} + \beta_3 \, \textit{Flood}_{i,t-1} + \beta_4 \, \mathbf{X}_{i,t} + \delta_i + \delta_t + \varepsilon_{i,t}, \quad (1)$$

where subscript i indexes the county and t the year. $Y_{i,t}$ is the log of annual county-level property tax revenue. $\textit{Treated}_{i,t-1}$ is the share of communities in county i with an effective flood map in the prior year. $\textit{Flood}_{i,t-1}$ is either a count of floods or the log of total flood-related property damage in the prior year. $\mathbf{X}_{i,t}$ includes the county-level employment rate, log income per capita, log population, and the share of properties at high flood risk. All controls vary over time except for the flood risk measure, which is based on projected exposure and is fixed over time. Standard errors are clustered at the county level. The fixed effects specifications allow me to exploit two sources of identifying variation. The first compares counties within the same state, year, and five-year mapping cohort, where cohorts are defined by the timing of initial flood map issuance. The second compares the same county over time using county and year fixed effects.

Table 2 shows that flood insurance access offsets the property tax revenue losses that follow floods. In the sample, real county property tax revenue grows by about 2% per year on average, providing a benchmark for the magnitude of the estimates. Conditional on state, cohort, and year fixed effects, a one standard deviation increase in prior-year flood events

[5] After Hurricane Irma, for example, Florida appraisers cut the assessed values of homes with unrepaired damage about three months after the storm (Figure C.2).

lowers annual property tax revenue by 1.3% in counties without insurance access, erasing about two-thirds of a typical year's growth. At the average level of county access, where about half of communities have access, flood insurance access more than offsets the estimated decline. The same conclusion holds when flood exposure is measured using flood-related property damage, indicating that household flood insurance stabilizes county property tax revenue by protecting the local tax base after floods.[6]

[Insert Table 2 Here]

These results are consistent with prior evidence that insurance payments help keep flooded households solvent and out of default (Gallagher and Hartley 2017) and finance repairs and reinvestment in damaged homes (McCoy and Zhao 2018). By supporting household recovery, insurance payments may also limit foreclosure spillovers that depress neighboring assessed values (Campbell, Giglio, and Pathak 2011).

7.2 Flood Insurance Access and Municipal Bond Markets

Flood insurance access can affect municipal borrowing costs through two opposing channels. The first is revenue stabilization. By helping households repair damaged property after floods, insurance can preserve the local tax base, reduce fiscal volatility, and lower the default risk priced by municipal bond investors. The second is flood risk exposure. By lowering the private cost of living in high-risk areas, subsidized insurance may encourage migration and infrastructure investment in vulnerable areas, increasing long-run fiscal risk. The property tax results in the preceding subsection provide direct evidence for the revenue stabilizing channel. The municipal bond tests examine whether investors price this revenue stabilization, net of any offsetting increase in future disaster exposure. I test this with the following specification:

$$Y_{i,t} = \alpha + \beta_1 \, \mathit{Treated}_{i,t} + \beta_2 \, \mathbf{X}_{i,t} + \delta_i + \delta_t + \varepsilon_{i,t}, \tag{2}$$

where subscript i indexes the community and t the month-year. $Y_{i,t}$ is the community-by-month-year average of the stated all-in cost. $\mathit{Treated}_{i,t}$ equals one if the community has received its initial flood map by the bond's sale date. $\mathbf{X}_{i,t}$ includes bond-level characteristics

[6]To examine how the effect evolves after a flood, I re-estimate the property tax specification at three annual horizons: the year of the flood, the next year, and the year after. I measure insurance access to match the start of each horizon, so that access is determined before the flood. Both the post-flood decline in property tax revenue and the offsetting effect of flood insurance access are largest in the year of the flood and attenuate over time (Table C.2). This timing is consistent with insurance stabilizing local revenues when flood-related losses are most pronounced. The offset is also larger where access is more widespread. When I replace the linear interaction with separate flood slopes estimated within four access bins, counties with broader access experience larger post-flood stabilization of property tax revenue (Figure C.3).

aggregated to the community-month level (maturity, amount, callable, credit enhancement, bank qualification, bid type, taxability) and county-level controls (employment rate, log income per capita, log population, flood risk share). Standard errors are clustered at the community level. Similar to Equation 1, I report specifications using either state, cohort, and month-year or community and month-year fixed effects.

The revenue stabilizing channel dominates. Access to subsidized flood insurance reduces stated all-in costs by about 4 basis points (Table 3, Columns 1–2). Because the within-community estimate absorbs time-invariant differences across communities, the result is unlikely to reflect pre-existing differences in borrowing costs. Borrowing costs fall when residents are able to purchase subsidized flood insurance.[7]

Issuance also increases after flood insurance access expands. I estimate the same specification using log total annual issuance at the community-year level, which better matches annual and multi-year municipal capital planning.[8] Total issuance rises by 3% after a community gains insurance access (Table 3, Columns 3–4).[9] Borrowing costs fall even as municipalities issue more debt. Holding investor demand fixed, greater bond supply would put downward pressure on prices and upward pressure on stated all-in costs. The fact that stated all-in costs fall while issuance rises suggests that this supply effect is more than offset by stronger investor demand for these bonds, consistent with flood insurance access improving perceived credit conditions.

[Insert Table 3 Here]

Because counties typically contain multiple NFIP communities, the community-level estimate measures the marginal effect of one community gaining access within a broader local

[7]In the Internet Appendix, I report additional tests showing that the main result is not driven by the timing or composition of flood map access. The coefficient on insurance access is essentially unchanged after dropping the earliest mapping cohort, which rules out the concern that the result is driven by permanent differences between early- and late-mapped communities (Table C.3, Column 1). The estimate also remains similar and statistically significant when I allow the effect of insurance access to vary with the share of community land in an SFHA (Table C.3, Columns 2–3). Because SFHAs are where floodplain building rules are strictest, the insignificant interaction between SFHA fraction and insurance access suggests that the main estimate is not driven by localized floodplain regulations. I also use randomization inference, which repeatedly assigns the observed flood map dates to different communities and compares the actual estimate to the distribution of estimates produced by these placebo assignments. The actual estimate falls in the extreme left tail of this placebo distribution, with p-values below 1 percent (Figure C.4, Panels A–B). Placebo access dates assigned one to three years before the actual flood map date produce no statistically significant effect, which helps rule out pre-existing trends in borrowing costs before access (Table C.4). Dynamic specifications show that stated all-in costs remain lower over the first decade after mapping, with no evidence of attenuation (Table C.5). This persistence is consistent with ongoing tax base stabilization rather than a short-lived repricing around flood map issuance.

[8]See Government Finance Officers Association (2023).

[9]Randomization inference on log issuance amount also rules out aggregate time trends as the source of the result (Figure C.4, Panels C–D).

market. When I measure access as the share of communities in a county with flood maps, full county access reduces the average stated all-in cost by about 17 basis points, or 3% of the unconditional sample mean of 5.49% (Table 4).

[Insert Table 4 Here]

The larger county-level estimate is consistent with spillovers across communities that share local housing markets and tax bases. The Callaway and Sant'Anna (2021) staggered difference-in-differences adjustment yields a similar 14 basis point decline once at least half of the communities in a county have been mapped (Figure 2), suggesting that treatment-effect heterogeneity in the two-way fixed effects design does not drive the result.

[Insert Figure 2 Here]

The flood insurance access estimate is similar in magnitude to common municipal bond pricing discounts, including those associated with bank qualification and credit enhancement at the county level.[10] Unlike these bond features, however, flood insurance access is not chosen or financed by the municipality. Bank qualification lowers yields through favorable tax treatment for eligible small issuers, and issuers may limit annual issuance to preserve eligibility (St. Clair 2024). Credit enhancement lowers yields through third-party repayment support that issuers typically purchase through premiums or fees. Flood insurance access instead lowers borrowing costs through a federally subsidized household insurance program outside of the bond contract. Because municipalities benefit from the reduction in borrowing costs without paying fees or limiting issuance, the estimated effect is a lower bound on the financial benefit to local governments.

7.3 Flood Insurance Access and Bond Markets After Floods

Insurance access should matter most when floods occur. I test this by interacting community-level access with realized flood exposure at the county level, using a version of Equation 1 estimated at the community-by-month-year level with the access variable lagged by 12 months.

Floods raise borrowing costs, but flood insurance access offsets most of the increase. Within a state and mapping cohort, a one standard deviation increase in the number of floods over the prior 12 months raises stated all-in costs by about 2 basis points, but flood insurance access offsets about 70% of this increase (Table 5, Column 1). The within-community estimate is similar, suggesting that mapped communities are largely insulated from increases

[10] Figure C.5 compares the access estimate to these benchmarks.

in borrowing costs after floods (Table 5, Column 2). The results also hold using log property damage as the measure of flood exposure (Table 5, Columns 3–4).[11]

[Insert Table 5 Here]

In the year following a flood, mapped communities increase issuance less than unmapped communities (Table 6). When insurance payouts help households repair damaged properties, they may limit declines in property values and preserve the local tax base. Unmapped communities, lacking this buffer, may be more likely to enter the bond market when floods weaken the local revenue base.

The borrowing cost and issuance results suggest that household flood insurance reduces local governments' perceived default risk. Mapped communities borrow less after floods and face smaller increases in borrowing costs when they do issue debt. By protecting property values and property tax revenues, flood insurance lowers municipalities' exposure to revenue risk. Investors appear to price insurance access as a source of local financial stability.

[Insert Table 6 Here]

7.4 Distributional Effects

Flood insurance access lowers borrowing costs on average, but the benefits may vary with actual insurance take-up. In 2018, FEMA reported that even within identical flood zones, higher-income households were more likely to enroll in NFIP coverage (FEMA 2018). If take-up is higher in wealthier counties, insurance access should provide greater protection to the local tax base and generate larger reductions in borrowing costs. Consistent with this mechanism, moving from the 25th to the 75th percentile of county income per capita is associated with an additional 4 basis point reduction in borrowing costs. This result suggests that subsidized flood insurance provides greater financial benefits to wealthier areas.

I also examine heterogeneity by flood exposure. Coastal counties, which typically have high flood risk, face stated all-in costs that are 12 basis points higher than inland counties on average. However, average levels of flood insurance access eliminate more than half of this premium (Table 7, Column 2). Last, densely populated counties may also benefit more from subsidized flood insurance because a single flood can disrupt a larger share of the tax base at once, magnifying the stabilizing effect on local finances. In counties with a one standard deviation increase in population density, full flood insurance access reduces stated all-in costs by about 7 basis points relative to counties without any access (Table 7, Column 3).

[11] I also use realized NFIP claim payouts to test whether actual insurance payments dampen the increase in borrowing costs after floods. They do. Floods raise stated all-in costs, but the increase is smaller in communities receiving larger payouts (Table C.6). Because claim payouts are post-flood transfers to insured households, this result suggests that actual insurance payments are reflected in municipal financing costs.

[Insert Table 7 Here]

The heterogeneity highlights an important distributional feature of the program. Insurance access is most valuable in bond markets where the protected tax base is large and where households are most likely to take up coverage. Wealthier counties are more likely to have higher NFIP take-up, while densely populated counties have more taxable property exposed within a given flood-prone area. As a result, the benefits that flow to local governments through lower borrowing costs are concentrated in wealthier, coastal, and densely populated counties. These results have important welfare implications. Although high flood risk counties may appear to need the most fiscal protection ex ante, lower borrowing costs can obscure the true extent of climate risk exposure, weaken incentives for local risk mitigation, and encourage continued investment in vulnerable areas. The program may therefore unintentionally reinforce existing inequalities while weakening the market signals that would otherwise discourage development in climate-vulnerable places.

8 Falsification Tests

Insurance access should affect borrowing costs only when the underlying shock works through the property tax base, and only when the insurance program provides coverage for the relevant weather-related event. Two placebo tests follow from this prediction. The first uses revenue bonds. Because these bonds are backed by project-specific cash flows, an insurance program that stabilizes property values should have little effect on their borrowing costs. The second uses wildfires. Because the NFIP does not cover wildfire damage, flood insurance access should not offset the increase in borrowing costs that follows a wildfire.

8.1 Revenue Bonds

In earlier sections, I find that stated all-in costs are lower in counties with greater access to federal flood insurance, which I argue reflects smoother property tax revenue. However, revenue bonds, unlike general obligation bonds, are secured by the cash flows of specific projects and should therefore be unaffected by property tax revenue. It follows that flood insurance access should not influence their pricing if the property tax mechanism drives the main results.[12]

The results show that flood insurance access does not affect revenue bonds' borrowing costs (Table 8, Columns 1–2). In addition, flood insurance access does not significantly reduce

[12]I exclude General Purpose and Water, Sewer, and Gas Facilities revenue bonds from the placebo sample. These bonds are directly exposed to flood risk because flood damage can affect system infrastructure, customer retention, and the revenues pledged to bondholders (Li, Saigal, and Adriaens 2024).

revenue bond borrowing costs following realized floods (Table 8, Columns 3–6). These results support the property tax mechanism. Flood insurance affects the bonds whose repayment most depends on property tax revenue, but does not produce a general reduction in flood-related credit risk across all municipal debt.

[Insert Table 8 Here]

8.2 Wildfires

I next test whether the flood results reflect a general capacity to withstand disasters or the specific role of flood insurance. Communities with flood insurance access may have greater administrative capacity, stronger infrastructure, or other characteristics that make them more resilient to weather shocks generally. If so, insurance access should also soften the effect of disasters that are not covered by the NFIP.

Wildfires provide a useful comparison to floods because, like floods, they are plausibly exogenous weather shocks that can disrupt local infrastructure. They damage property, are increasing in frequency and severity, and are priced in municipal credit markets (Berry-Stölzle and Hao 2026; Liao and Kousky 2022). Unlike floods, however, wildfire damage is not covered by the NFIP. If flood insurance access proxies for general disaster resilience, then counties with greater access should also face smaller increases in borrowing costs after wildfires.

They do not. Wildfires increase borrowing costs, but the interaction between flood insurance access and wildfires is small and statistically insignificant (Table 9).[13] These placebo results suggest that the main estimates reflect the specific role of flood insurance in reducing flood-related repayment risk.

[Insert Table 9 Here]

9 Conclusion

A growing literature documents the effects of floods and flood risk on household financial outcomes, house prices, and municipal borrowing. To my knowledge, this is the first paper to examine whether expanding residents' eligibility to purchase subsidized flood insurance lowers municipal financing costs. In this paper, I first show that expanding households' access to flood insurance stabilizes property tax revenues. Second, I show that stated all-in

[13]Because wildfires occur infrequently in the data, I also include snowstorms as a second placebo test. Snowstorms are the second most frequent weather shock in the SHELDUS data after floods and cause about half as much median property damage as floods. Snow damage is also not covered by the NFIP. The interaction between flood insurance access and snowstorms is statistically insignificant as well (Table C.7).

costs increase after a flood, but flood insurance access partially offsets the increase. Third, I provide evidence that areas where residents hedge their exposure to flood risk have lower financing costs. These results show that, in addition to benefiting individuals, flood insurance access has substantial effects on the fiscal health of the municipalities in which these individuals reside.

Overall, this paper finds that social welfare programs designed to help individuals can have effects beyond direct recipients, including effects on municipal bond markets. Floods are not the only natural disaster increasing in frequency in recent decades; wildfires, tornadoes, and other extreme weather events have also become more frequent and more damaging. As a result, private insurance companies have become more reluctant to offer natural disaster coverage, restricting access or raising rates in many areas of the country (Horn and Webel 2024). If the federal government fills new coverage gaps for these other types of natural disasters as it has done for floods, there may be additional spillover effects on local governments. This paper emphasizes the need to account for broader fiscal externalities when designing federal policy.

Figure 1. Staggered Rollout of Initial Flood Maps: Bergen County, New Jersey

This figure illustrates the staggered rollout of initial flood maps across National Flood Insurance Program (NFIP) communities within a representative county. Each polygon corresponds to an NFIP community in Bergen County, New Jersey, shaded by the year in which its initial flood map became effective, with darker shades indicating later mapping dates. Unincorporated areas are portions of the county that lie outside the corporate limits of any incorporated city or town.

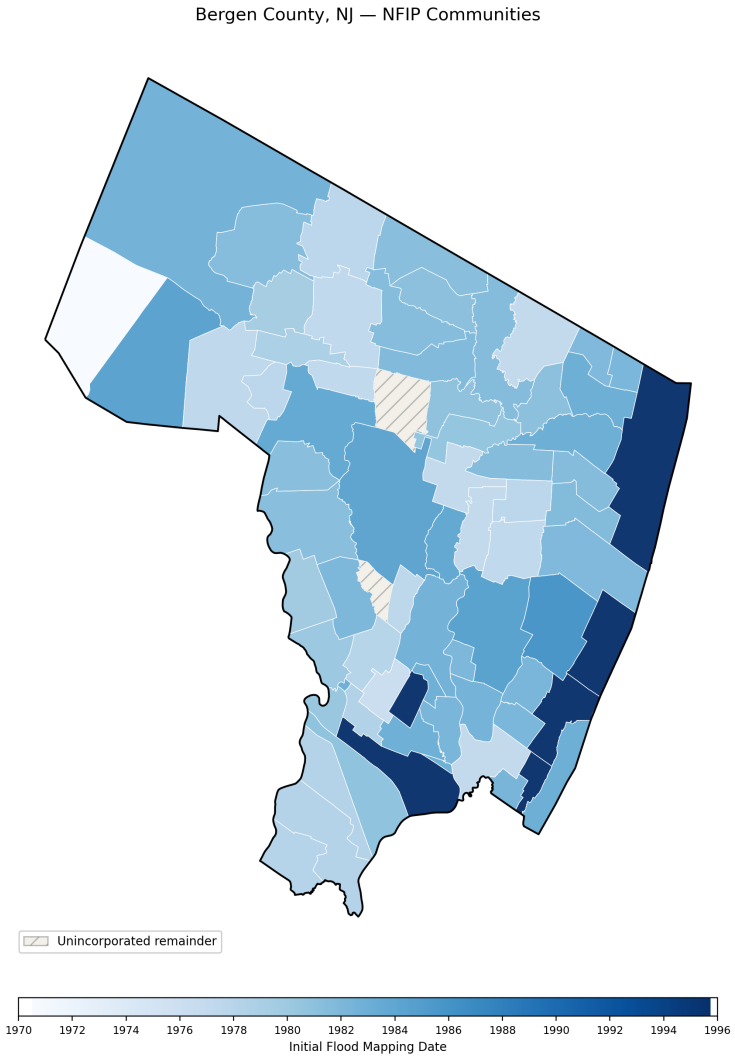


Figure 2. Callaway and Sant'Anna (2021) Adjustment: Effect of Flood Insurance Access on Stated All-in Cost

This figure presents the results in Section 7 using the approach proposed by Callaway and Sant'Anna (2021) to correct the biases of the two-way fixed effects estimates of difference-in-differences coefficients in the specification: $Y_{i,t} = \alpha + \beta_1 Treated_{i,t} + \delta_i + \delta_t + \varepsilon_{i,t}$. Subscript i represents the county and subscript t represents the year. The observations are at the county-by-year level. The sample is restricted to general obligation bonds with maturities greater than one year and a stated all-in cost below 50 percent, if populated. The outcome variable is $Stated\ All\text{-}in\ Cost_{i,t}$, the average annual cost of the debt, including underwriting and other fees. $Treated_{i,t}$ is an indicator variable equal to one if at least half of the communities within a county have an effective flood map by the bond sale date, and zero otherwise. Coefficient estimates are plotted with 95% confidence intervals based on standard errors clustered at the county level.

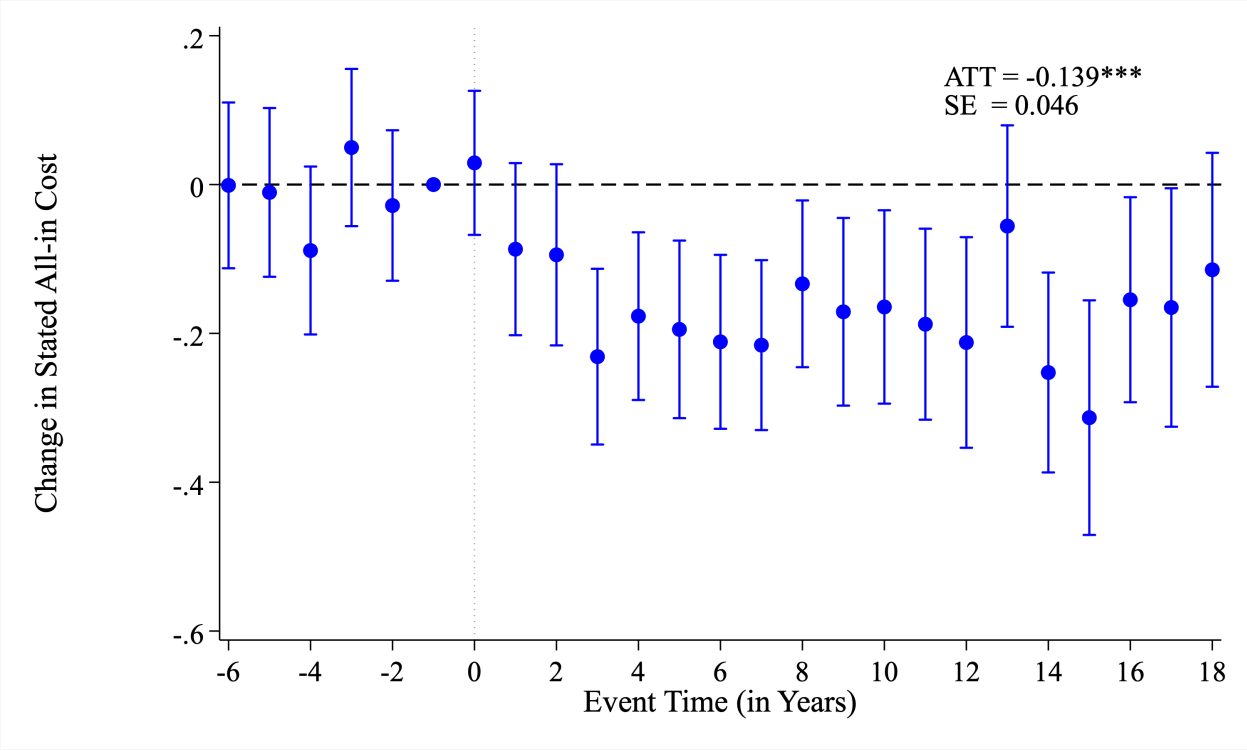


Table 1. Summary Statistics of General Obligation Bonds at the Community Level

This table reports summary statistics for community-level outcomes, treatment variables, and covariates. The sample includes all National Flood Insurance Program (NFIP) communities matched to general obligation bond issuers between 1969 and 2019. The sample is restricted to general obligation bonds with maturities greater than one year and a stated all-in cost below 50 percent, if populated. Bond-level data are collapsed to the community-by-month-year level, yielding 78,763 community-by-month-year observations of 9,641 unique communities. Subscript i represents the community and subscript t represents the month-year. $Stated\ All\text{-}in\ Cost_{i,t}$ is the average annual cost of debt issued by community i in month t, including underwriting and other fees. $Log\ Total\ Annual\ Issuance\ Amount_{i,t}$ is the log of the total dollar amount of bonds issued by the community annually. $Flood\ Insurance\ Access_{i,t}$ is a binary indicator equal to one if the community's initial flood map date is on or before the bond sale date, and zero otherwise. $Flood\ Insurance\ Access_{i,t-1}$ is the same indicator measured 12 months before the sale date. $Issuance\ Amount_{i,t}$ and $Years\ to\ Maturity_{i,t}$ are community-level monthly averages of bond issuance amounts and years to maturity. $Taxable_{i,t}$, $Credit\ Enhanced_{i,t}$, $Bank\ Qualified_{i,t}$, $Negotiated_{i,t}$, and $Callable_{i,t}$ are monthly community-level averages of indicator variables equal to one if the bond is taxable, credit-enhanced, bank-qualified, negotiated, or callable, respectively, and zero otherwise. $Flood_{i,t-1}$ is a count of the number of floods in the community's county in the prior 12 months. $Flood\ Damage_{i,t-1}$ is the CPI-adjusted property damage caused by floods in the community's county over the same period, in millions of dollars. $Snowstorm_{i,t-1}$ is the analogous count of snowstorms in the community's county in the prior 12 months. $Wildfire_{i,t-1}$ is the analogous count of wildfires in the community's county in the prior 12 months. $Monthly\ Precipitation_{i,t}$ and $Monthly\ Temperature_{i,t}$ are the county-level monthly mean precipitation in millimeters and mean temperature in degrees Celsius. $High\ Flood\ Risk_i$ is the share of properties in the community's county projected to face high-to-severe flood risk between 2022 and 2052 from First Street Foundation. $SFHA\ Fraction_i$ is the share of the community in a Special Flood Hazard Area, computed from FEMA's National Flood Hazard Layer. $Income\ Per\ Capita_{i,t}$ and $Employment\ Rate_{i,t}$ are annual county-level measures from the U.S. Bureau of Economic Analysis (BEA). $Coast_i$ is an indicator equal to one if the community's county is on the coast and zero otherwise. $Population\ Density_{i,t}$ is the county's annual population divided by its 1990 land area.

Variable	Mean	SD	P10	P50	P90	N
Dependent Variables						
Stated All-in Cost$_{i,t}$ (%)	5.485	1.973	3.083	5.338	7.813	78,763
Log Total Annual Issuance Amount$_{i,t}$	0.900	1.590	−1.124	0.894	2.890	60,575
Treatment Variables						
Flood Insurance Access$_{i,t}$	0.628	0.483	0.000	1.000	1.000	78,763
Flood Insurance Access$_{i,t-1}$	0.608	0.488	0.000	1.000	1.000	78,763
Bond Characteristics						
Issuance Amount$_{i,t}$ ($ MM)	5.313	17.838	0.295	2.000	10.420	78,763
Years to Maturity$_{i,t}$	14.940	9.007	5.000	15.000	23.000	78,763
Taxable$_{i,t}$	0.017	0.123	0.000	0.000	0.000	78,763
Credit Enhanced$_{i,t}$	0.181	0.380	0.000	0.000	1.000	78,763
Bank Qualified$_{i,t}$	0.383	0.482	0.000	0.000	1.000	78,763
Negotiated$_{i,t}$	0.873	0.332	0.000	1.000	1.000	78,763
Callable$_{i,t}$	0.405	0.487	0.000	0.000	1.000	78,763
Flood and Climate Variables						
Flood$_{i,t-1}$ (prior 12 months)	1.955	1.896	0.000	1.000	5.000	78,763
Flood Damage$_{i,t-1}$ ($ MM)	10.983	245.625	0.000	0.021	2.057	78,763
Snowstorm$_{i,t-1}$ (prior 12 months)	0.835	1.290	0.000	0.000	3.000	78,763
Wildfire$_{i,t-1}$ (prior 12 months)	0.012	0.134	0.000	0.000	0.000	78,763
Monthly Precipitation$_{i,t}$ (mm)	85.351	61.824	18.298	75.131	160.783	78,738
Monthly Temperature$_{i,t}$ (°C)	12.408	9.736	−1.505	13.742	23.957	78,738
Community and County Demographics						
High Flood Risk$_i$	0.122	0.093	0.056	0.102	0.191	78,763
Income Per Capita$_{i,t}$ ($)	38,287	11,896	26,303	36,106	52,663	78,763
Employment Rate$_{i,t}$	0.508	0.124	0.360	0.495	0.671	78,763
SFHA Fraction$_i$	0.105	0.118	0.000	0.074	0.232	78,763
Coast$_i$	0.141	0.348	0.000	0.000	1.000	78,763
Population Density$_{i,t}$	0.286	0.722	0.009	0.074	0.743	78,763

Table 2. Does Flood Insurance Access Stabilize Property Tax Revenue After a Flood?

This table presents OLS estimates of the following staggered difference-in-differences tests at the county level. The specifications are: $Y_{i,t} = \alpha + \beta_1 \, Treated_{i,t-1} \times Flood_{i,t-1} + \beta_2 \, Treated_{i,t-1} + \beta_3 \, Flood_{i,t-1} + \beta_4 \mathbf{X}_{i,t} + \delta_m + \delta_c + \delta_t + \varepsilon_{i,t}$ and $Y_{i,t} = \alpha + \beta_1 \, Treated_{i,t-1} \times Flood_{i,t-1} + \beta_2 \, Treated_{i,t-1} + \beta_3 \, Flood_{i,t-1} + \beta_4 \mathbf{X}_{i,t} + \delta_i + \delta_t + \varepsilon_{i,t}$. Subscript i represents the county, subscript m represents the state, subscript c represents the five-year FEMA mapping cohort, and subscript t represents the year. The observations are at the county-by-year level. The outcome variable is *Log Property Tax Revenue*$_{i,t}$, the log of annual county-level property tax revenue. $Treated_{i,t-1}$ is *Flood Insurance Access*$_{i,t-1}$, the share of communities in the county with an effective flood map as of the prior year. In Columns 1–2, $Flood_{i,t-1}$ is the standardized count of floods in the county over the prior year. In Columns 3–4, $Flood_{i,t-1}$ is *Log Flood Damage*$_{i,t-1}$, the log of total flood-related property damage in the county over the same period. *High Flood Risk*$_i$ is constructed from First Street Foundation's model. *Log Income Per Capita*$_{i,t}$, *Employment Rate*$_{i,t}$, and *Log Population*$_{i,t}$ are annual county-level measures from the U.S. Bureau of Economic Analysis (BEA). Robust standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	Log Property Tax Revenue			
	(1)	(2)	(3)	(4)
Flood Insurance Access$_{i,t-1}$	-0.055***	-0.004	-0.066***	-0.009
	(0.021)	(0.016)	(0.022)	(0.017)
Flood$_{i,t-1}$	-0.013**	-0.007		
	(0.005)	(0.005)		
Flood$_{i,t-1}$× Flood Insurance Access$_{i,t-1}$	0.056***	0.039***		
	(0.008)	(0.006)		
Log Flood Damage$_{i,t-1}$			-0.003***	-0.001***
			(0.001)	(0.001)
Log Flood Damage$_{i,t-1}$× Flood Insurance Access$_{i,t-1}$			0.009***	0.006***
			(0.001)	(0.001)
High Flood Risk$_i$	0.443***		0.440***	
	(0.078)		(0.077)	
Employment Rate$_{i,t}$	0.336***	-0.210**	0.336***	-0.213**
	(0.073)	(0.085)	(0.074)	(0.086)
Log Population$_{i,t}$	0.840***	1.070***	0.842***	1.071***
	(0.008)	(0.022)	(0.008)	(0.022)
Log Income Per Capita$_{i,t}$	0.864***	0.489***	0.864***	0.490***
	(0.045)	(0.035)	(0.045)	(0.036)
Observations	101,460	107,600	101,460	107,600
R-squared	0.902	0.949	0.902	0.948
Year FE	Yes	Yes	Yes	Yes
State & Cohort FE	Yes	No	Yes	No
County FE	No	Yes	No	Yes
Number of Counties	2,809	3,031	2,809	3,031
Mean of Dependent Variable	9.066	8.992	9.066	8.992

Table 3. Effect of Flood Insurance Access on Borrowing Costs and Amounts

This table presents OLS estimates of the following staggered difference-in-differences tests. The specifications are: $Y_{i,t} = \alpha + \beta_1 Treated_{i,t} + \beta_2 \mathbf{X}_{i,t} + \delta_m + \delta_c + \delta_t + \varepsilon_{i,t}$ and $Y_{i,t} = \alpha + \beta_1 Treated_{i,t} + \beta_2 \mathbf{X}_{i,t} + \delta_i + \delta_t + \varepsilon_{i,t}$. Subscript i represents the community, subscript m represents the state, subscript c represents the five-year FEMA mapping cohort, and subscript t represents the month-year (Columns 1–2) or year (Columns 3–4). The sample is collapsed to the community-by-month-year level for the stated all-in cost regressions and to the community-by-year level for the issuance amount regressions. The sample is restricted to general obligation bonds with maturities greater than one year and a stated all-in cost below 50 percent, if populated. In Columns 1–2, the outcome variable is *Stated All-in Cost*$_{i,t}$, the average annual cost of debt the issuer pays on the bond, including underwriting and other fees. In Columns 3–4, the outcome variable is *Log Issuance Amount*$_{i,t}$, the log of the total issuance amount at the community level each year. $Treated_{i,t}$ is a binary variable equal to *Flood Insurance Access*$_{i,t}$, one if the community has received its initial flood map by the bond sale date, and zero otherwise. $Negotiated_{i,t}$, $Taxable_{i,t}$, *Credit Enhanced*$_{i,t}$, $Callable_{i,t}$, and *Bank Qualified*$_{i,t}$ are monthly community-level averages of indicator variables for whether the bond is negotiated, taxable, credit enhanced, callable, and bank qualified. When included as controls, *Log Maturity*$_{i,t}$ and *Log Issuance Amount*$_{i,t}$ are the log of community-level monthly averages of maturities and issuance amounts. *High Flood Risk*$_i$ is the share of properties in the community's county projected to face high-to-severe flood risk between 2022 and 2052 from First Street Foundation. *Log Income Per Capita*$_{i,t}$, *Employment Rate*$_{i,t}$, and *Log Population*$_{i,t}$ are annual county-level measures from the BEA. Robust standard errors clustered at the community level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	Stated All-in Cost		Log Issuance Amount	
	(1)	(2)	(3)	(4)
Flood Insurance Access$_{i,t}$	-0.037**	-0.036**	0.033*	0.032**
	(0.016)	(0.016)	(0.018)	(0.015)
Negotiated$_{i,t}$	-0.238***	-0.197***	0.044**	-0.075***
	(0.014)	(0.013)	(0.018)	(0.011)
Taxable$_{i,t}$	1.148***	1.149***	-0.250***	-0.181***
	(0.034)	(0.032)	(0.034)	(0.025)
Credit Enhanced$_{i,t}$	-0.066***	-0.109***	0.652***	0.525***
	(0.009)	(0.011)	(0.013)	(0.009)
Callable$_{i,t}$	0.118***	0.121***	0.393***	0.360***
	(0.010)	(0.009)	(0.015)	(0.010)
Bank Qualified$_{i,t}$	-0.063***	-0.084***	-1.131***	-0.598***
	(0.010)	(0.009)	(0.018)	(0.011)
Log Maturity$_{i,t}$	0.620***	0.585***	0.261***	0.285***
	(0.010)	(0.009)	(0.010)	(0.007)
Log Issuance Amount$_{i,t}$	-0.066***	-0.042***		
	(0.005)	(0.004)		
High Flood Risk$_{i}$	0.267***		-0.200	
	(0.071)		(0.166)	
Log Income Per Capita$_{i,t}$	-0.076*	-0.201***	0.288***	0.384***
	(0.044)	(0.060)	(0.079)	(0.108)
Employment Rate$_{i,t}$	-0.319***	-0.058	0.281**	0.543***
	(0.076)	(0.112)	(0.131)	(0.130)
Log Population$_{i,t}$	0.021***	-0.062**	0.252***	0.587***
	(0.007)	(0.027)	(0.012)	(0.048)
Observations	76,989	76,555	135,700	137,839
R-squared	0.914	0.933	0.486	0.732
Month-Year FE	Yes	Yes	No	No
Year FE	No	No	Yes	Yes
State & Cohort FE	Yes	No	Yes	No
Community FE	No	Yes	No	Yes
Number of Communities	9,254	7,433	11,719	10,649
Mean of Dependent Variable	5.487	5.474	1.375	1.365

Table 4. County-Level Specification: Effect of Flood Insurance Access

This table presents OLS estimates of the following staggered difference-in-differences tests at the county level. The specifications are: $Y_{i,t} = \alpha + \beta_1 Treated_{i,t} + \beta_2 \mathbf{X}_{i,t} + \delta_m + \delta_c + \delta_t + \varepsilon_{i,t}$ and $Y_{i,t} = \alpha + \beta_1 Treated_{i,t} + \beta_2 \mathbf{X}_{i,t} + \delta_i + \delta_t + \varepsilon_{i,t}$. Subscript i represents the county, subscript m represents the state, subscript c represents the five-year FEMA mapping cohort, and subscript t represents the month-year (Columns 1–2) or year (Columns 3–4). The sample is collapsed to the county-by-month-year level for the stated all-in cost regressions and to the county-by-year level for the issuance amount regressions. The sample is restricted to general obligation bonds with maturities greater than one year and a stated all-in cost below 50 percent, if populated. In Columns 1–2, the outcome variable is *Stated All-in Cost*$_{i,t}$, the average annual cost of debt the issuer pays on the bond, including underwriting and other fees. In Columns 3–4, the outcome variable is *Log Issuance Amount*$_{i,t}$, the log of the total issuance amount at the county level each year. $Treated_{i,t}$ is *Flood Insurance Access*$_{i,t}$, the share of communities in the county whose initial flood map had been issued by the sale date. Bond and county control variables are the same as those in Table 3. Robust standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	Stated All-in Cost		Log Issuance Amount	
	(1)	(2)	(3)	(4)
Flood Insurance Access$_{i,t}$	-0.061**	-0.170***	0.026	0.028
	(0.025)	(0.029)	(0.039)	(0.035)
Observations	66,155	67,484	61,880	63,760
R-squared	0.918	0.926	0.694	0.752
Month-Year FE	Yes	Yes	No	No
Year FE	No	No	Yes	Yes
Bond & County Controls	Yes	Yes	Yes	Yes
State & Cohort FE	Yes	No	Yes	No
County FE	No	Yes	No	Yes
Number of Counties	2,506	2,340	2,734	2,718
Mean of Dependent Variable	5.498	5.494	1.869	1.829

Table 5. Effect of Floods and Flood Insurance Access on Stated All-in Cost

This table presents OLS estimates of the following staggered difference-in-differences tests. The specifications are: $Y_{i,t} = \alpha + \beta_1 \, Treated_{i,t-1} \times Flood_{i,t-1} + \beta_2 \, Treated_{i,t-1} + \beta_3 \, Flood_{i,t-1} + \beta_4 \mathbf{X}_{i,t} + \delta_m + \delta_c + \delta_t + \varepsilon_{i,t}$ and $Y_{i,t} = \alpha + \beta_1 \, Treated_{i,t-1} \times Flood_{i,t-1} + \beta_2 \, Treated_{i,t-1} + \beta_3 \, Flood_{i,t-1} + \beta_4 \mathbf{X}_{i,t} + \delta_i + \delta_t + \varepsilon_{i,t}$. Subscript i represents the community, subscript m represents the state, subscript c represents the five-year FEMA mapping cohort, and subscript t represents the month-year. The observations are at the community-by-month-year level. The sample is restricted to general obligation bonds with maturities greater than one year and a stated all-in cost below 50 percent, if populated. The outcome variable is *Stated All-in Cost*$_{i,t}$, the average annual cost of debt the issuer pays on the bond, including underwriting and other fees. $Treated_{i,t-1}$ is a binary variable representing *Flood Insurance Access*$_{i,t-1}$, one if the community had received its initial flood map 12 months before the sale date and zero otherwise. In Columns 1–2, $Flood_{i,t-1}$ is the standardized count variable for the number of floods in the community's county in the prior 12 months. In Columns 3–4, *Log Flood Damage*$_{i,t-1}$ is the log of total flood-related property damage in the community's county over the same period. Bond and county control variables are the same as those in Table 3. Robust standard errors clustered at the community level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	Stated All-in Cost			
	(1)	(2)	(3)	(4)
Flood Insurance Access$_{i,t-1}$	-0.032*	-0.026	-0.039**	-0.029
	(0.017)	(0.018)	(0.016)	(0.018)
Flood$_{i,t-1}$	0.021***	0.014***		
	(0.005)	(0.005)		
Flood$_{i,t-1}$× Flood Insurance Access$_{i,t-1}$	-0.015**	-0.020***		
	(0.006)	(0.007)		
Log Flood Damage$_{i,t-1}$			0.002**	0.002**
			(0.001)	(0.001)
Log Flood Damage$_{i,t-1}$× Access$_{i,t-1}$			-0.001	-0.002**
			(0.001)	(0.001)
Observations	76,989	76,555	76,989	76,555
R-squared	0.915	0.933	0.914	0.933
Month-Year FE	Yes	Yes	Yes	Yes
Bond & County Controls	Yes	Yes	Yes	Yes
State & Cohort FE	Yes	No	Yes	No
Community FE	No	Yes	No	Yes
Number of Communities	9,254	7,433	9,254	7,433
Mean of Dependent Variable	5.487	5.474	5.487	5.474

Table 6. Effect of Floods and Flood Insurance Access on Log Issuance Amount

This table presents OLS estimates of the following staggered difference-in-differences tests. The specifications are: $Y_{i,t} = \alpha + \beta_1 Treated_{i,t-1} \times Flood_{i,t-1} + \beta_2 Treated_{i,t-1} + \beta_3 Flood_{i,t-1} + \beta_4 \mathbf{X}_{i,t} + \delta_m + \delta_c + \delta_t + \varepsilon_{i,t}$ and $Y_{i,t} = \alpha + \beta_1 Treated_{i,t-1} \times Flood_{i,t-1} + \beta_2 Treated_{i,t-1} + \beta_3 Flood_{i,t-1} + \beta_4 \mathbf{X}_{i,t} + \delta_i + \delta_t + \varepsilon_{i,t}$. Subscript i represents the community, subscript m represents the state, subscript c represents the five-year FEMA mapping cohort, and subscript t represents the year. The observations are at the community-by-year level. The sample is restricted to general obligation bonds with maturities greater than one year and a stated all-in cost below 50 percent, if populated. The outcome variable is $Log\ Issuance\ Amount_{i,t}$, the log of the total issuance amount at the community level each year. $Treated_{i,t-1}$ is a binary variable representing $Flood\ Insurance\ Access_{i,t-1}$, one if the community had received its initial flood map 12 months before the sale date and zero otherwise. In Columns 1–2, $Flood_{i,t-1}$ is the standardized count variable for the number of floods in the community's county in the prior 12 months. In Columns 3–4, $Log\ Flood\ Damage_{i,t-1}$ is the log of total flood-related property damage in the community's county over the same period. Bond and county control variables are the same as those in Table 3, Columns 3–4. Robust standard errors clustered at the community level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	Log Issuance Amount			
	(1)	(2)	(3)	(4)
Flood Insurance Access$_{i,t-1}$	0.073***	0.051***	0.088***	0.040**
	(0.021)	(0.017)	(0.023)	(0.018)
Flood$_{i,t-1}$	0.032***	0.025***		
	(0.010)	(0.008)		
Flood$_{i,t-1}$× Flood Insurance Access$_{i,t-1}$	-0.057***	-0.031***		
	(0.012)	(0.008)		
Log Flood Damage$_{i,t-1}$			0.006***	0.002
			(0.002)	(0.001)
Log Flood Damage$_{i,t-1}$× Access$_{i,t-1}$			-0.009***	-0.002
			(0.002)	(0.001)
Observations	135,700	137,839	135,700	137,839
R-squared	0.486	0.732	0.486	0.732
Year FE	Yes	Yes	Yes	Yes
Bond & County Controls	Yes	Yes	Yes	Yes
State & Cohort FE	Yes	No	Yes	No
Community FE	No	Yes	No	Yes
Number of Communities	11,719	10,649	11,719	10,649
Mean of Dependent Variable	1.375	1.365	1.375	1.365

Table 7. Heterogeneous Effects of Flood Insurance Access on Stated All-in Cost

This table presents OLS estimates of staggered difference-in-differences tests based on heterogeneous effects. The specification is: $Y_{i,t} = \alpha + \beta_1 \, Treated_{i,t} \times Z_{i,t} + \beta_2 \, Treated_{i,t} + \beta_3 \, Z_{i,t} + \beta_4 \mathbf{X}_{i,t} + \delta_m + \delta_c + \delta_t + \varepsilon_{i,t}$. Subscript i represents the county, subscript m represents the state, subscript c represents the five-year FEMA mapping cohort, and subscript t represents the month-year. The observations are at the county-by-month-year level. The sample is restricted to general obligation bonds with maturities greater than one year and a stated all-in cost below 50 percent, if populated. The outcome variable is *Stated All-in Cost*$_{i,t}$, the average annual cost of debt the issuer pays on the bond, including underwriting and other fees. *Treated*$_{i,t}$ is *Flood Insurance Access*$_{i,t}$, the share of communities in the county whose initial flood map had been issued. In Column 1, $Z_{i,t}$ is *Log Income Per Capita*$_{i,t}$. In Column 2, Z_i is *Coast*$_i$, an indicator for whether the county is on the coast. In Column 3, $Z_{i,t}$ is *Population Density*$_{i,t}$, defined as the county population divided by the county's 1990 land area, standardized. Control variables are the same as those in Table 3, except the log of income per capita is omitted in Column 1 and the log of population is omitted in Column 3 to avoid collinearity with the heterogeneity variable. Robust standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	Stated All-in Cost		
	(1)	(2)	(3)
Flood Insurance Access$_{i,t}$	1.187**	-0.040	-0.019
	(0.481)	(0.025)	(0.026)
Log Income Per Capita$_{i,t}$	0.007	-0.062	-0.044
	(0.053)	(0.055)	(0.050)
Flood Insurance Access$_{i,t}$× Log Income Per Capita$_{i,t}$	-0.119***		
	(0.046)		
Coast$_i$		0.119***	
		(0.039)	
Flood Insurance Access$_{i,t}$× Coast$_i$		-0.155***	
		(0.033)	
Population Density$_{i,t}$			0.087***
			(0.019)
Flood Insurance Access$_{i,t}$× Population Density$_{i,t}$			-0.066***
			(0.019)
Observations	66,155	66,155	66,155
R-squared	0.918	0.918	0.918
Month-Year FE	Yes	Yes	Yes
Bond & County Controls	Yes	Yes	Yes
State & Cohort FE	Yes	Yes	Yes
Number of Counties	2,506	2,506	2,506
Mean of Dependent Variable	5.498	5.498	5.498

Table 8. Effect of Flood Insurance Access on Revenue Bonds

This table presents OLS estimates of staggered difference-in-differences tests for revenue bonds. The specifications are: $Y_{i,t} = \alpha + \beta_1 Treated_{i,t} + \beta_2 \mathbf{X}_{i,t} + \delta_m + \delta_c + \delta_t + \varepsilon_{i,t}$, $Y_{i,t} = \alpha + \beta_1 Treated_{i,t} + \beta_2 \mathbf{X}_{i,t} + \delta_i + \delta_t + \varepsilon_{i,t}$, and $Y_{i,t} = \alpha + \beta_1 Treated_{i,t-1} \times Flood_{i,t-1} + \beta_2 Treated_{i,t-1} + \beta_3 Flood_{i,t-1} + \beta_4 \mathbf{X}_{i,t} + \delta_i + \delta_t + \varepsilon_{i,t}$. Subscript i represents the community, subscript m represents the state, subscript c represents the five-year FEMA mapping cohort, and subscript t represents the month-year. The observations are at the community-by-month-year level. The sample is restricted to revenue bonds with maturities greater than one year and a stated all-in cost below 50 percent. The sample is further restricted to exclude Water, Sewer, and Gas Facilities and General Purpose bonds. The outcome variable is *Stated All-in Cost*$_{i,t}$, the average annual cost of debt the issuer pays on the bond, including underwriting and other fees. $Treated_{i,t-1}$ is a binary variable representing *Flood Insurance Access*$_{i,t-1}$, one if the community had received its initial flood map 12 months before the sale date and zero otherwise. In Columns 3–4, $Flood_{i,t-1}$ is the standardized count variable for the number of floods in the community's county in the prior 12 months. In Columns 5–6, *Log Flood Damage*$_{i,t-1}$ is the log of total flood-related property damage in the community's county over the same period. Bond and county control variables are the same as those in Table 3. Robust standard errors clustered at the community level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	Stated All-in Cost					
	(1)	(2)	(3)	(4)	(5)	(6)
Flood Insurance Access$_{i,t}$	-0.031	-0.002				
	(0.027)	(0.030)				
Flood Insurance Access$_{i,t-1}$			-0.032	0.011	-0.060*	-0.022
			(0.031)	(0.035)	(0.032)	(0.036)
Flood$_{i,t-1}$			-0.001	0.008		
			(0.012)	(0.013)		
Flood$_{i,t-1}$× Flood Insurance Access$_{i,t-1}$			-0.003	-0.010		
			(0.014)	(0.016)		
Log Flood Damage$_{i,t-1}$					-0.002	-0.001
					(0.002)	(0.002)
Log Flood Damage$_{i,t-1}$× Access$_{i,t-1}$					0.003	0.003
					(0.002)	(0.003)
Observations	19,385	17,924	19,385	17,924	19,385	17,924
R-squared	0.898	0.922	0.898	0.922	0.898	0.922
Month-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Bond & County Controls	Yes	Yes	Yes	Yes	Yes	Yes
State & Cohort FE	Yes	No	Yes	No	Yes	No
Community FE	No	Yes	No	Yes	No	Yes
Number of Communities	4,056	2,449	4,056	2,449	4,056	2,449
Mean of Dependent Variable	6.886	6.848	6.886	6.848	6.886	6.848

Table 9. Effect of Wildfires on Stated All-in Cost

This table presents OLS estimates of staggered difference-in-differences tests for the interaction of flood insurance access with wildfires. The specifications are: $Y_{i,t} = \alpha + \beta_1 Treated_{i,t-1} \times Wildfire_{i,t-1} + \beta_2 Treated_{i,t-1} + \beta_3 Wildfire_{i,t-1} + \beta_4 \mathbf{X}_{i,t} + \delta_m + \delta_c + \delta_t + \varepsilon_{i,t}$ and $Y_{i,t} = \alpha + \beta_1 Treated_{i,t-1} \times Wildfire_{i,t-1} + \beta_2 Treated_{i,t-1} + \beta_3 Wildfire_{i,t-1} + \beta_4 \mathbf{X}_{i,t} + \delta_i + \delta_t + \varepsilon_{i,t}$. Subscript i represents the community, subscript m represents the state, subscript c represents the five-year FEMA mapping cohort, and subscript t represents the month-year. The observations are at the community-by-month-year level. The sample is restricted to general obligation bonds with maturities greater than one year and a stated all-in cost below 50 percent, if populated. The outcome variable is *Stated All-in Cost*$_{i,t}$, the average annual cost of debt the issuer pays on the bond, including underwriting and other fees. $Treated_{i,t-1}$ is a binary variable representing *Flood Insurance Access*$_{i,t-1}$, one if the community had received its initial flood map 12 months before the sale date and zero otherwise. In Columns 1–2, $Wildfire_{i,t-1}$ is the standardized count variable for the number of wildfires in the community's county in the prior 12 months. In Columns 3–4, *Log Fire Damage*$_{i,t-1}$ is the log of total wildfire-related property damage in the community's county over the same period. Bond and county control variables are the same as those in Table 3. Robust standard errors clustered at the community level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	Stated All-in Cost			
	(1)	(2)	(3)	(4)
Flood Insurance Access$_{i,t-1}$	-0.046***	-0.044***	-0.046***	-0.045***
	(0.016)	(0.016)	(0.016)	(0.016)
Wildfire$_{i,t-1}$	0.013**	0.012**		
	(0.007)	(0.006)		
Wildfire$_{i,t-1}\times$ Flood Insurance Access$_{i,t-1}$	-0.007	-0.006		
	(0.007)	(0.006)		
Log Fire Damage$_{i,t-1}$			0.007*	0.007**
			(0.004)	(0.003)
Log Fire Damage$_{i,t-1}\times$ Access$_{i,t-1}$			-0.002	-0.002
			(0.004)	(0.004)
Observations	76,989	76,555	76,989	76,555
R-squared	0.914	0.933	0.914	0.933
Month-Year FE	Yes	Yes	Yes	Yes
Bond & County Controls	Yes	Yes	Yes	Yes
State & Cohort FE	Yes	No	Yes	No
Community FE	No	Yes	No	Yes
Number of Communities	9,254	7,433	9,254	7,433
Mean of Dependent Variable	5.487	5.474	5.487	5.474

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Internet Appendix

A Conceptual Framework

This appendix develops the conceptual framework introduced in Section 3. The framework formalizes a simple bond pricing intuition. General obligation bonds are backed by local tax revenue. Floods can decrease revenue by damaging taxable property, and flood insurance can stabilize revenue by helping households repair damaged structures and remain in place. Holding development fixed, insurance access lowers municipal yields through two terms. First, it reduces expected flood losses to the local tax base. Second, it reduces the volatility of those losses. The final subsection relaxes the assumption that development is held fixed and separates the revenue stabilizing channel from a potential moral hazard channel.

The appendix proceeds in four steps. Section A.1 summarizes how NFIP access enters the framework. Section A.2 derives expected flood losses and their volatility and prices the bond through a mean-variance marginal investor's optimality condition. Section A.3 derives comparative statics and connects them to the empirical heterogeneity tests. Section A.4 separates the revenue stabilizing channel from the moral hazard channel.

I price the bond using the marginal condition of a mean-variance investor. Another approach would be to use a structural model, such as Merton (1974) (see also Goldsmith-Pinkham et al. (2023)). Although I believe such an approach would yield similar conclusions, the specifications would differ in how payoff dispersion enters pricing. In the structural model, dispersion matters through the concavity of the debt payoff, which is present even under risk-neutral pricing. In the mean-variance framework it matters because the marginal investor requires compensation for bearing undiversified local repayment risk.

A.1 Primitives and the Institutional Structure of Access

Consider community i. Let F_i be a measure over taxable properties, so that $dF_i(h)$ counts properties with value h. Aggregate taxable value before a flood is

$$B_i = \int h \, dF_i(h),$$

where h is the value of an individual taxable property.

Not all taxable property is equally exposed to flood risk. Let

$$E_i = s_i B_i$$

denote taxable value exposed to flood risk, where $s_i \in [0,1]$ is the share of the local tax base exposed to flood losses. Equivalently, let F_i^E denote the corresponding measure over damageable values among exposed properties, so that $E_i = \int h \, dF_i^E(h) = s_i B_i$. In the model, E_i should be read as the damageable component of exposed taxable value. The effective coverage term below maps residential insurance payouts into preservation of the broader property tax base, rather than assuming that every component of taxable value is directly insured.

A flood occurs with probability π_i. Conditional on a flood, the event damages a fraction L of exposed structure value. Because the local government relies partly on property taxation, flood damage affects public revenues in proportion to the exposed tax base.

NFIP access enters through insurance eligibility. Let $a_i \in \{0,1\}$ equal one once a community participates in the NFIP and has a finalized flood map. After access, eligible households can purchase NFIP coverage, and some properties face mandatory purchase requirements if they are located in the Special Flood Hazard Area (SFHA) and financed with federally backed mortgages. The resulting coverage, θ_i^*, is the share of exposed taxable value effectively insured once access is available. This parameter captures the combined effect of all coverage determinants, including income, mortgage prevalence, risk salience, and prior flood experience.

NFIP building coverage is capped at C. For an insured exposed property of value h, flood damage is Lh, the insurance payout is $\min(Lh, C)$, and the residual uncovered loss is $\max(Lh - C, 0)$. Define the coverage adequacy ratio over exposed property as

$$\kappa_i \equiv \frac{\int \min(Lh, C) \, dF_i^E(h)}{\int Lh \, dF_i^E(h)} = \frac{\int \min(Lh, C) \, dF_i^E(h)}{LE_i}.$$

The parameter $\kappa_i \in [0,1]$ measures the fraction of gross flood damage covered by insurance among exposed properties. If the cap never binds, then $\kappa_i = 1$. If the cap binds, then $\kappa_i < 1$. The product $\theta_i^* \kappa_i$ should be read as the share of physical damage that does not translate into a persistent decline in property taxable value because insurance payouts finance repair, rebuilding, or continued occupancy.

A.2 Expected Flood Losses, Loss Volatility, and Municipal Yields

A community issues a general obligation bond at date 0 that promises one dollar of repayment at date 1, and the bond trades today at price p_i. Between issuance and repayment a flood occurs with probability π_i, indicated by $Z_i \in \{0,1\}$ with $P(Z_i = 1) = \pi_i$. The flood is the only source of risk to repayment, and it either has or has not occurred by the time the bond pays.

If a flood occurs, uninsured exposed property loses a fraction L of value and insured exposed property retains the share that payouts allow households to repair, so the residual proportional damage to exposed value is

$$\ell_i(a_i) = L(1 - a_i\theta_i^*\kappa_i).$$

Without access, $a_i = 0$ and the residual damage is L. With access, full take-up, and a non-binding cap, $a_i\theta_i^*\kappa_i = 1$, so the residual damage is zero. Partial stabilization leaves it between zero and L. The residual damage measures the loss in property value that remains after repair, not the physical damage that insurance prevents, because payouts finance the repair, rebuilding, and continued occupancy that keep flood damage from becoming a persistent decline in the property tax base.

A flood reduces the property base by the fraction $s_i\ell_i(a_i)$, the exposed share times the residual damage. Let $\beta_i \geq 0$ denote the reduced form pass-through from a proportional decline in the property tax base to the priced credit loss per dollar promised. Intuitively, β_i measures how much of a shock to the tax base reaches bondholders as priced risk. It is smaller where a community can absorb the shock without straining repayment, such as through reserves, support from the state, or a strong legal claim on revenue, and larger where it cannot.

This pass-through is consistent with municipal credit analysis. General obligation credit is not priced as a mechanical ratio of promised repayment to assessed property value. Rating methodologies evaluate the economy and tax base, reserves, management and institutions, legal pledge strength, and debt and pension burdens (Moody's Investors Service 2021). The parameter β_i summarizes how these features, together with the property tax rate, determine the extent to which a proportional shock to the tax base is reflected in bond prices and yields. I hold β_i fixed in the comparative statics because the model studies how insurance changes residual flood exposure, not municipal leverage.

The bondholder's loss per dollar promised is the pass-through times the proportional base loss,

$$\Lambda_i(a_i) = \beta_i\, s_i\, \ell_i(a_i)\, Z_i = \beta_i s_i L(1 - a_i\theta_i^*\kappa_i)\, Z_i,$$

a flood indicator scaled by the residual exposure. Because Z_i is Bernoulli, the first two moments are

$$\mathbb{E}[\Lambda_i(a_i)] = \beta_i s_i L(1 - a_i\theta_i^*\kappa_i)\,\pi_i, \qquad \mathrm{Var}(\Lambda_i(a_i)) = \big(\beta_i s_i L\big)^2(1 - a_i\theta_i^*\kappa_i)^2\,\pi_i(1 - \pi_i).$$

Holding exposed development fixed, insurance access lowers both moments by raising $a_i\theta_i^*\kappa_i$,

the share of exposed flood losses that insurance offsets. I normalize effective pre-access coverage to zero, so the comparison is between $a_i = 1$ and $a_i = 0$.

The bond's payoff per dollar promised is $x_i(a_i) = 1 - \Lambda_i(a_i)$, equal to one if no flood occurs and $1 - \beta_i s_i \ell_i(a_i)$ if one does. Its price is the expected payoff under the marginal investor's stochastic discount factor M_i,

$$p_i = \mathbb{E}[M_i \, x_i(a_i)] \, .$$

The municipal market is segmented along tax and geographic lines, because the tax exemption concentrates holdings among in-state and retail investors who are imperfectly diversified against local repayment risk (Pirinsky and Wang 2011; Ang et al. 2010; Poterba 1989). For a marginal investor with mean-variance preferences, the discount factor is affine and decreasing in the return on the portfolio she holds, $M_i = \eta - b \, R_{p,i}$ with $b > 0$. Combining $p_i = \mathbb{E}[M_i x_i]$ with the corresponding condition for the risk-free asset, and writing the gross return as $R_i = x_i/p_i$, gives the standard pricing relation

$$\mathbb{E}[R_i] = \frac{1}{\mathbb{E}[M_i]} + \gamma \, \mathrm{Cov}(R_{p,i}, R_i), \qquad \gamma \equiv \frac{b}{\mathbb{E}[M_i]} > 0,$$

where $1/\mathbb{E}[M_i] = 1 + r_f + \chi_i$ is the gross return on a riskless tax-exempt claim and χ_i is the wedge from the taxable risk-free rate created by the tax exemption, liquidity, and maturity. An asset earns a premium only for the part of its risk that covaries with the investor's portfolio. A diversified investor would hold little exposure to any single community, and the local flood, being idiosyncratic to the broad economy, would carry almost no premium. The segmented investor instead holds bonds exposed to the same regional floods, so her portfolio return moves with this bond's flood loss and $\mathrm{Cov}(R_{p,i}, R_i) \approx \mathrm{Var}(R_i)$. The flood shock is therefore priced through the variance the investor bears, and the marginal condition is

$$\mathbb{E}[\tilde{R}_i] = 1 + r_f + \chi_i + \gamma \, \mathrm{Var}(\tilde{R}_i).$$

The yield y_i is the rate that discounts the promised dollar to the current price, $p_i = 1/(1 + y_i)$. Substituting the return and the loss into the marginal condition and expanding around small losses gives[14]

$$y_i(a_i) = r_f + \chi_i + \mu \, \mathbb{E}[\Lambda_i(a_i)] + \lambda \, \mathrm{Var}(\Lambda_i(a_i)), \qquad \mu = 1 + r_f + \chi_i, \qquad \lambda = \gamma(1 + r_f + \chi_i)^2.$$

[14]Writing $R_i = (1 - \Lambda_i)/p_i$ and $1 + y_i = 1/p_i$, the marginal condition becomes $(1 - \mathbb{E}[\Lambda_i])(1 + y_i) = 1 + r_f + \chi_i + \gamma \, \mathrm{Var}(\Lambda_i)(1 + y_i)^2$. Expanding to first order in $\mathbb{E}[\Lambda_i]$ and $\mathrm{Var}(\Lambda_i)$ around the no-loss yield $1 + y_i = 1 + r_f + \chi_i$ gives the stated expression.

The level weight μ is the gross required return and the risk weight λ is effective risk aversion scaled by that return. Both are positive. I refer to $\mu\,\mathbb{E}[\Lambda_i(a_i)]$ as the level term and $\lambda\,\mathrm{Var}(\Lambda_i(a_i))$ as the volatility term. The empirical analysis absorbs the variation in the weights through fixed effects and bond controls, so I suppress their community subscript.

Substituting the two moments gives the pricing equation

$$y_i(a_i) = r_f + \chi_i + \mu\,\pi_i\,\beta_i Ls_i(1 - a_i\theta_i^*\kappa_i) + \lambda\,\pi_i(1-\pi_i)\big[\beta_i Ls_i(1 - a_i\theta_i^*\kappa_i)\big]^2.$$

Holding exposed development fixed, this shows why insurance access lowers yields when it raises the share of exposed flood losses that insurance offsets. The level term falls because insurance access reduces expected residual damage to the base, and the volatility term falls because insurance access reduces the sensitivity of the base to flood realizations. The same comparative statics apply to any increase in this share after access, not only to the onset of NFIP coverage.

The yield reduction from insurance access, $\Delta y_i \equiv y_i(0) - y_i(1)$, is

$$\Delta y_i = \underbrace{\mu\,\pi_i\,\beta_i Ls_i\,\theta_i^*\kappa_i}_{\text{level}} + \underbrace{\lambda\,\pi_i(1-\pi_i)(\beta_i Ls_i)^2\theta_i^*\kappa_i(2-\theta_i^*\kappa_i)}_{\text{volatility}}.$$

The level term reflects the reduction in expected uninsured damage, and the volatility term reflects the reduction in uncertainty. Both terms are larger when flood probability, the exposed share of the tax base, the fiscal pass-through to bond prices, and effective coverage are higher.

A.3 Implications for Heterogeneity

The pricing equation delivers three empirical predictions, which guide the heterogeneity tests in the empirical analysis. First, the reduction in borrowing costs from insurance access is larger when effective insurance coverage is higher. Taking the derivative of the yield reduction, Δy_i, with respect to effective coverage, θ_i^*, gives

$$\frac{\partial \Delta y_i}{\partial \theta_i^*} = \mu\,\pi_i\,\beta_i Ls_i\,\kappa_i + 2\lambda\,\pi_i(1-\pi_i)(\beta_i Ls_i)^2\kappa_i(1-\theta_i^*\kappa_i) > 0.$$

The first term is the level effect, because greater coverage reduces expected uninsured damage. The second term is the volatility effect, because greater coverage reduces the sensitivity of the exposed tax base to flood realizations. I use income as an empirical proxy for effective coverage, because high-income households are more likely to purchase and maintain flood insurance (FEMA 2018). The model therefore predicts larger reductions in borrowing costs

in high-income communities. This prediction can weaken at the very top of the property value distribution, but holds on average.

Second, the reduction in borrowing costs from insurance access is larger when flood probability is higher. Differentiating the yield reduction from insurance access with respect to π_i gives

$$\frac{\partial \Delta y_i}{\partial \pi_i} = \mu\, \beta_i L s_i\, \theta_i^* \kappa_i + \lambda(1 - 2\pi_i)(\beta_i L s_i)^2 \theta_i^* \kappa_i (2 - \theta_i^* \kappa_i).$$

This derivative is positive over the empirically relevant range of annual flood probabilities. The level term rises linearly in flood probability because expected uninsured damage scales with the chance of a flood. The volatility term rises over the relevant range because flood events generate larger swings in the exposed tax base when they are more common but not nearly certain. In the empirical analysis, coastal counties and realized flood exposure proxy for π_i.

Third, the reduction in borrowing costs from insurance access is larger when the exposed share of the tax base is high. Differentiating the yield reduction from insurance access with respect to the exposed share gives

$$\frac{\partial \Delta y_i}{\partial s_i} = \mu\, \pi_i\, \beta_i L\, \theta_i^* \kappa_i + 2\lambda\, \pi_i (1 - \pi_i)(\beta_i L)^2 s_i \theta_i^* \kappa_i (2 - \theta_i^* \kappa_i) > 0.$$

The yield reduction from insurance access rises with the exposed share, linearly through the level term and quadratically through the volatility term. I proxy for exposure concentration with density, because conditional on floodplain presence, denser places are more likely to have substantial taxable structures in exposed areas.

A.4 The Revenue Stabilizing Channel and the Moral Hazard Channel

The derivation so far holds the exposed share fixed and treats insurance access as binary. To study how insurance availability can change behavior, I make the subsidy component of insurance access explicit. With endogenous location and development decisions, subsidized flood insurance can affect yields through two opposing channels. On the one hand, subsidies increase coverage, which reduces residual flood losses and stabilizes the tax base. On the other hand, subsidies may make exposed locations more attractive, raising the share of taxable value placed in flood-prone areas.

Let ζ denote the subsidy rate, or the fraction of the true risk-based price that the program waives, so the subsidized premium equals $(1 - \zeta)$ times the actuarially fair premium. Let the exposed share and effective coverage both depend on the subsidy rate, $s_i(\zeta)$ and $\theta_i^*(\zeta)$.

Holding π_i, L, κ_i, and β_i fixed, the level and volatility terms depend on

$$s_i(\zeta)\left(1-\theta_i^*(\zeta)\kappa_i\right).$$

Note I hold κ_i, the share of flood damage the insurance payout covers, fixed with respect to the subsidy rate, so that subsidies work only by changing how much taxable value is insured, $\theta_i^*(\zeta)$, and how much sits in flood-prone areas, $s_i(\zeta)$. The level term is proportional to this product and the volatility term to its square, so the same condition determines whether the subsidy lowers both expected losses and loss volatility.

Taking logs and differentiating with respect to the subsidy rate,

$$\frac{\partial \ln\left[s_i(1-\theta_i^*\kappa_i)\right]}{\partial \zeta} = \frac{\partial \ln s_i}{\partial \zeta} - \frac{\kappa_i}{1-\theta_i^*\kappa_i}\frac{\partial \theta_i^*}{\partial \zeta}.$$

The first term is the moral hazard channel, because subsidies can raise the exposed share by encouraging development of flood-prone land. The second term is the revenue stabilizing channel, because subsidies raise coverage and reduce residual losses to the tax base. Insurance access lowers yields when the revenue stabilizing channel dominates,

$$\frac{\kappa_i}{1-\theta_i^*\kappa_i}\frac{\partial \theta_i^*}{\partial \zeta} > \frac{\partial \ln s_i}{\partial \zeta}.$$

The empirical finding that insurance access lowers stated all-in costs is consistent with the revenue stabilizing channel dominating on average in the part of municipal repayment risk reflected in yields.

The within-community estimates in Section 7 provide further evidence on this point. These estimates compare a community to itself before and after access, absorbing fixed differences in baseline exposure, local public finances, and development patterns. The comparative statics and the empirical sign are consistent with insurance access lowering yields by stabilizing the priced component of municipal repayment risk.

B Variable Definitions and Data Construction

This appendix provides additional details on data construction beyond the summary in Section 5.

B.1 Municipal Bond Issuance Data

Municipal bond data come from the Securities Data Company (SDC) Platinum Global Public Finance U.S. New Issues database. The full database contains 552,270 issuance-level observations covering 53,060 unique community-state-county combinations between 1966 and 2019. The data report bond characteristics including rating, issuance amount, final maturity, embedded options, credit enhancement, taxability, issuer county, sale method, and bank qualification status. I restrict the main sample to general obligation bonds issued between 1969 and 2019. I require bonds to have a place identifier, a reported issuance amount, and non-missing bond characteristics. I drop observations with a stated all-in cost above 50 percent, which are likely data errors, and I drop bonds with maturities of one year or less, because fluctuations in property values are more likely to affect the pricing of longer-term debt. The maturity restriction reduces the sample by about 10%.

SDC defines the stated all-in cost as the average annual cost of debt the issuer pays on the bond, including underwriting and other fees. It is the most consistently reported measure of borrowing costs in SDC before 1990, when initial yields and true interest cost are reported for fewer than 200 county-month observations per year. In months when both are reported, the stated all-in cost is 90% correlated with initial yields. Revenue bonds, which are backed by project-specific cash flows rather than the general tax base, are excluded from the main sample and used as a placebo test in Section 8.

B.2 Matching Bond Issuers to NFIP Communities

The community-level analysis requires matching each bond issuer in SDC to a specific NFIP Community Identifier (CID). FEMA's Community Status Book lists 24,866 NFIP-participating communities across 56 states and territories, each with its own initial flood map date. I match SDC issuers to CSB communities in three stages. First, I use exact name matches between the geographic component of the issuer name and the parsed community name within the same state and county. Second, when an exact match is not found, I apply fuzzy string matching to unmatched issuers, requiring a minimum similarity score of 85%. Third, issuers that cannot be matched to a specific city, town, or village are assigned to the unincorporated county CID.

The final match rate is 96.5%, with 51,221 of 53,060 unique issuer-state-county combinations linked to a CID. Of the matched issuers, 59% link to a specific incorporated community with its own flood map date. Another 38% are county-wide entities such as school districts and special authorities that are assigned the county's overall insurance access status. For robustness, I also run the main results on a sample that is restricted to city-matched communities only (Table C.1).

B.3 Community-Level Flood Insurance Data

The Community Status Book provides the date each community first received a flood map. I use the initial flood map date to construct the main treatment variable. A community is classified as having insurance access once it has received and accepted its initial flood map. Communities that received but rejected their maps are excluded. At the community level, the treatment indicator is equal to one if the community's flood map was issued on or before the bond sale date.

For county-level outcomes and aggregate specifications, I construct a continuous measure of insurance access as the equal-weighted share of communities within a county with an effective flood map by a given date, following Peralta and Scott (2024). Because a single NFIP community can span more than one county, I first assign each community identifier to a single primary county, defined as the county containing the largest share of the community's residents. I measure residents using Census sub-county population estimates, mapping each Census place to the county that holds the largest share of its population, and assign the community to that county's five-digit Federal Information Processing Standards (FIPS) code. The continuous insurance access measure is then keyed to the county FIPS code by month-year, and county fiscal and demographic data are merged on the same code. This measure reflects eligibility to purchase subsidized NFIP coverage rather than actual take-up.

I supplement the Community Status Book with two additional FEMA sources. First, I obtain community-level NFIP claims data from FEMA's OpenFEMA API, covering 2.7 million claims across 15,543 communities between 1978 and 2019. I use these data to construct rolling 12-month measures of claim frequency and CPI-adjusted claim payouts. Second, I compute community-level Special Flood Hazard Area shares by spatially intersecting FEMA flood-zone polygons with Census TIGER/Line place boundaries, which I use to classify communities into flood-risk quartiles.

B.4 County Fiscal Outcomes and Demographics

Property tax revenue comes from the U.S. Census Bureau's Annual Survey of State and Local Government Finances (Pierson et al. 2015). I adjust all revenue figures to 2020 dollars

using the Consumer Price Index from the U.S. Bureau of Labor Statistics. Property tax revenue is observed for all counties every five years, with annual reporting available for a subset of larger counties.

County-level population, per capita income, and employment come from the Bureau of Economic Analysis and are available annually beginning in 1969. I use the BEA employment rate rather than the unemployment rate because it is available throughout the full sample period, while county-level unemployment rates are only available beginning in 1990 (Deryugina 2017). I construct an indicator for whether a county's per capita income is above the national median in a given year. Population density is the county's annual population divided by its 1990 land area, as reported by the U.S. Census Bureau. The coastal indicator is taken from the NOAA coastal counties list.

B.5 Flood and Weather Event Data

County-level weather event data come from the Spatial Hazard Events and Losses Database, maintained by the Center for Emergency Management and Homeland Security. SHELDUS reports event dates, affected counties, event types, and inflation-adjusted property damage estimates. I use flood events between 1969 and 2019 that caused positive property damage. This restriction removes about 3% of flood observations and focuses the analysis on events with measurable local damage. I construct two county-month flood exposure variables: a count of floods in the prior 12 months, and the log of total flood-related property damage over the same period. I apply the same construction to wildfires and to snowstorms, classified as Wildfire and Winter Weather in SHELDUS, respectively. Both serve as falsification tests because neither wildfire nor snow damage is covered by the NFIP. Monthly precipitation and temperature controls come from the PRISM Climate Group at Oregon State University. Both variables enter in levels and squares to allow for nonlinear weather effects on local fiscal conditions.

B.6 Merging and Sample Construction

For the community-level analysis, each bond observation is linked to its matched CID and assigned flood insurance access based on whether the community's initial flood map date precedes the bond sale date. County-level flood events are merged according to whether a flood occurred within the 12 months before the issuance month. For the property tax revenue analysis, I match each county to the share of communities with an effective flood map as of 12 months before the end of its fiscal year. This lag reduces concerns about contemporaneous feedback between mapping and fiscal outcomes. For the dynamic specification reported in Table C.2, I construct flood exposure and flood insurance access at horizons of zero through

two years relative to the fiscal year. For each horizon $k \in \{0, 1, 2\}$, flood exposure is the standardized flood count (or the log of flood damage) over the 12 months ending $12k$ months before the end of the fiscal year, and the matching flood insurance access measure is the share of communities with an effective flood map as of $12(k + 1)$ months before the end of the fiscal year. Measuring insurance access one year ahead of each flood horizon ensures that insurance access is observed strictly before the corresponding flood window and is therefore unaffected by any post-flood take-up of insurance. All county characteristics are merged annually by calendar year.

The main bond market analysis aggregates bond-level observations to the community-by-month-year level using equal-weighted averages across bonds issued by the same community in the same month. This aggregation prevents communities that issue multiple bonds in a single month from mechanically receiving more weight in the analysis. Results are similar when the regression is weighted so that each community contributes equally, regardless of how frequently it issues debt.

C Additional Figures and Tables

Figure C.1. Relationship Between Initial Yields and Stated All-in Cost

This figure plots the relationship between initial yields and stated all-in cost between 1969 and 2019. The sample contains 65,590 county-by-month-year observations. The initial yield is the county-level average initial yield in a given month. The stated all-in cost is the county-level average annual cost of debt issued in a given month, including underwriting and other fees. The correlation between the two measures in the sample is 90%.

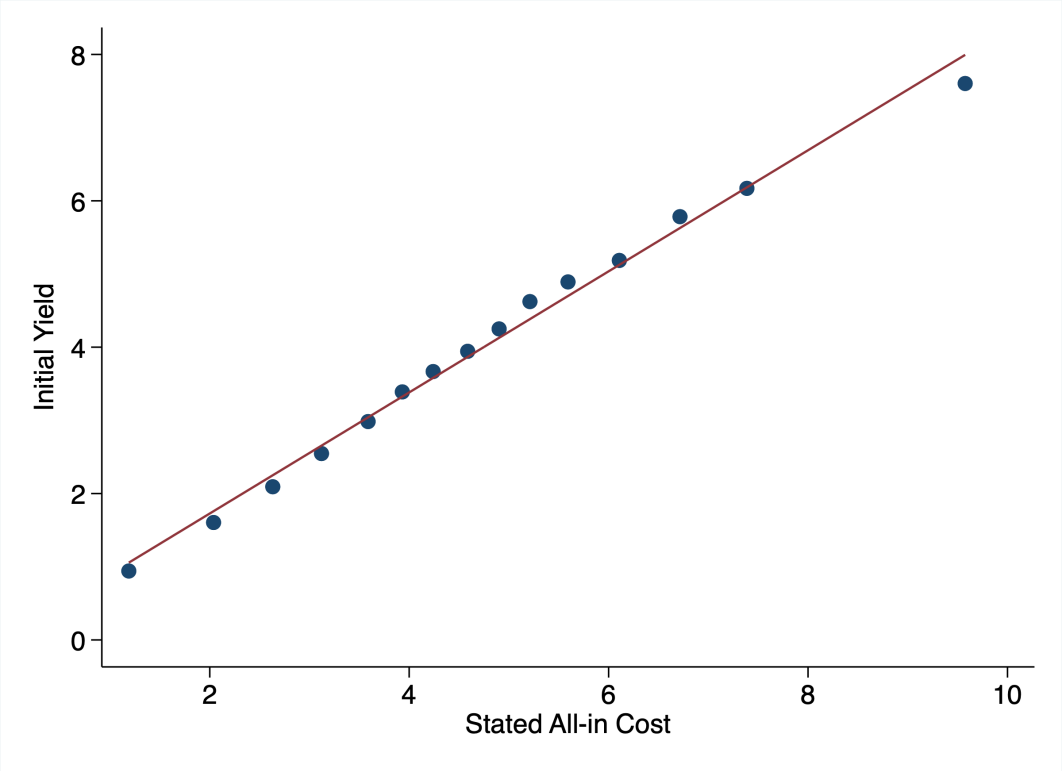
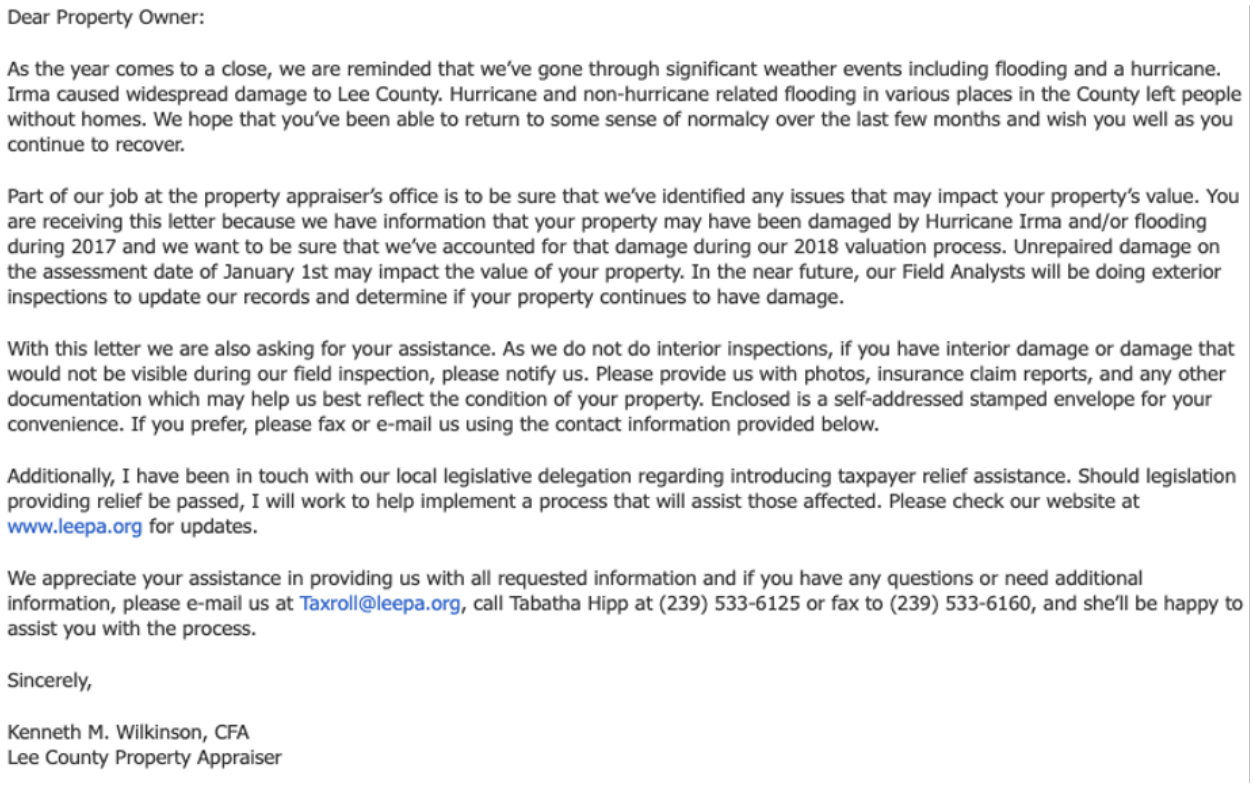


Figure C.2. Example of Property Reappraisal after a Hurricane

This figure shows a reappraisal notice sent to Florida residents on December 11, 2017, three months after Hurricane Irma made landfall. Local property appraisers automatically adjusted assessed values for homes with unrepaired damage.[15]



Dear Property Owner:

As the year comes to a close, we are reminded that we've gone through significant weather events including flooding and a hurricane. Irma caused widespread damage to Lee County. Hurricane and non-hurricane related flooding in various places in the County left people without homes. We hope that you've been able to return to some sense of normalcy over the last few months and wish you well as you continue to recover.

Part of our job at the property appraiser's office is to be sure that we've identified any issues that may impact your property's value. You are receiving this letter because we have information that your property may have been damaged by Hurricane Irma and/or flooding during 2017 and we want to be sure that we've accounted for that damage during our 2018 valuation process. Unrepaired damage on the assessment date of January 1st may impact the value of your property. In the near future, our Field Analysts will be doing exterior inspections to update our records and determine if your property continues to have damage.

With this letter we are also asking for your assistance. As we do not do interior inspections, if you have interior damage or damage that would not be visible during our field inspection, please notify us. Please provide us with photos, insurance claim reports, and any other documentation which may help us best reflect the condition of your property. Enclosed is a self-addressed stamped envelope for your convenience. If you prefer, please fax or e-mail us using the contact information provided below.

Additionally, I have been in touch with our local legislative delegation regarding introducing taxpayer relief assistance. Should legislation providing relief be passed, I will work to help implement a process that will assist those affected. Please check our website at www.leepa.org for updates.

We appreciate your assistance in providing us with all requested information and if you have any questions or need additional information, please e-mail us at Taxroll@leepa.org, call Tabatha Hipp at (239) 533-6125 or fax to (239) 533-6160, and she'll be happy to assist you with the process.

Sincerely,

Kenneth M. Wilkinson, CFA
Lee County Property Appraiser

Figure C.3. Effect of Floods on Property Tax Revenue by Level of Flood Insurance Access

This figure plots the marginal effect of floods on county property tax revenue at each level of flood insurance access. The estimates come from a county fixed effects regression of the log of annual county property tax revenue on $Flood_{i,t-1}$, the standardized count of flood events in the county in the prior year, interacted with indicators for the level of $Flood\ Insurance\ Access_{i,t-1}$, the share of communities in the county with an effective flood map in the prior year. I partition $Flood\ Insurance\ Access_{i,t-1}$ into four bins: $[0, 0.25)$, $[0.25, 0.5)$, $[0.5, 0.75)$, and $[0.75, 1]$. Each bar is the within-bin coefficient on the interaction, so that it measures the effect of a one standard deviation increase in flood events on log property tax revenue at that level of access. The regression includes the same county control variables as Table 2, along with year and county fixed effects. Capped vertical lines denote 95 percent confidence intervals based on standard errors clustered at the county level.

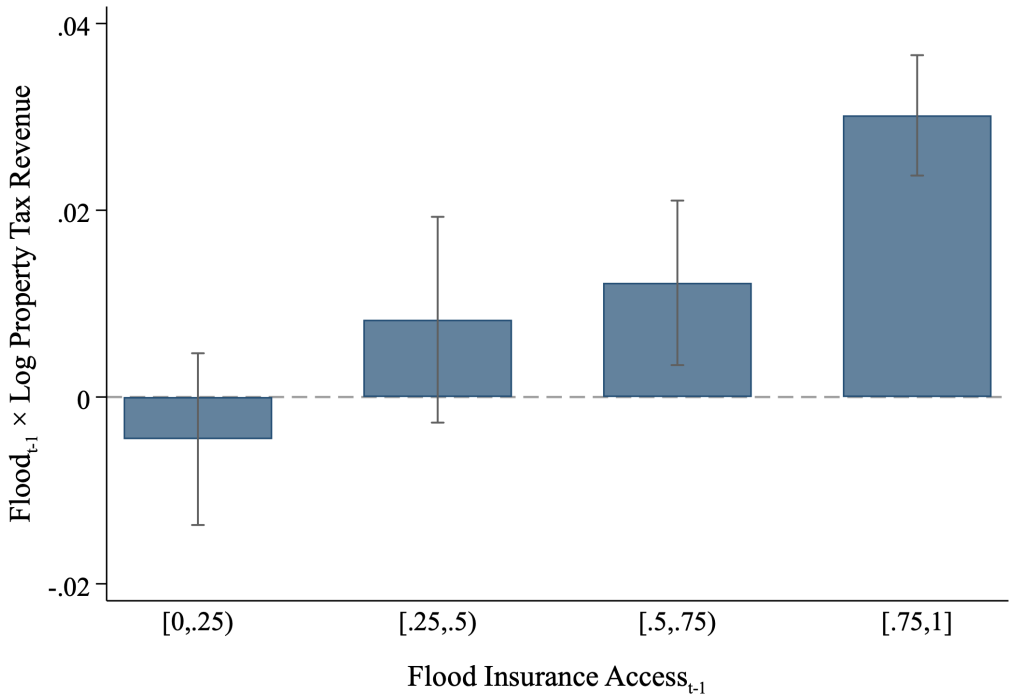


Figure C.4. Distribution of Placebo Coefficients on Borrowing Costs and Issuance Amounts

This figure plots the distribution of placebo coefficients on $Flood\ Insurance\ Access_{i,t}$ from 1,000 randomization inference iterations on $Stated\ All\text{-}in\ Cost_{i,t}$ (Panels A and B) and $Log\ Issuance\ Amount_{i,t}$ (Panels C and D). In each iteration, flood map dates are permuted across communities and the main specifications (matching Table 3) are re-estimated. Panels A and C report placebo coefficients from the state and cohort fixed effects specification. Panels B and D report placebo coefficients from the community fixed effects specification.

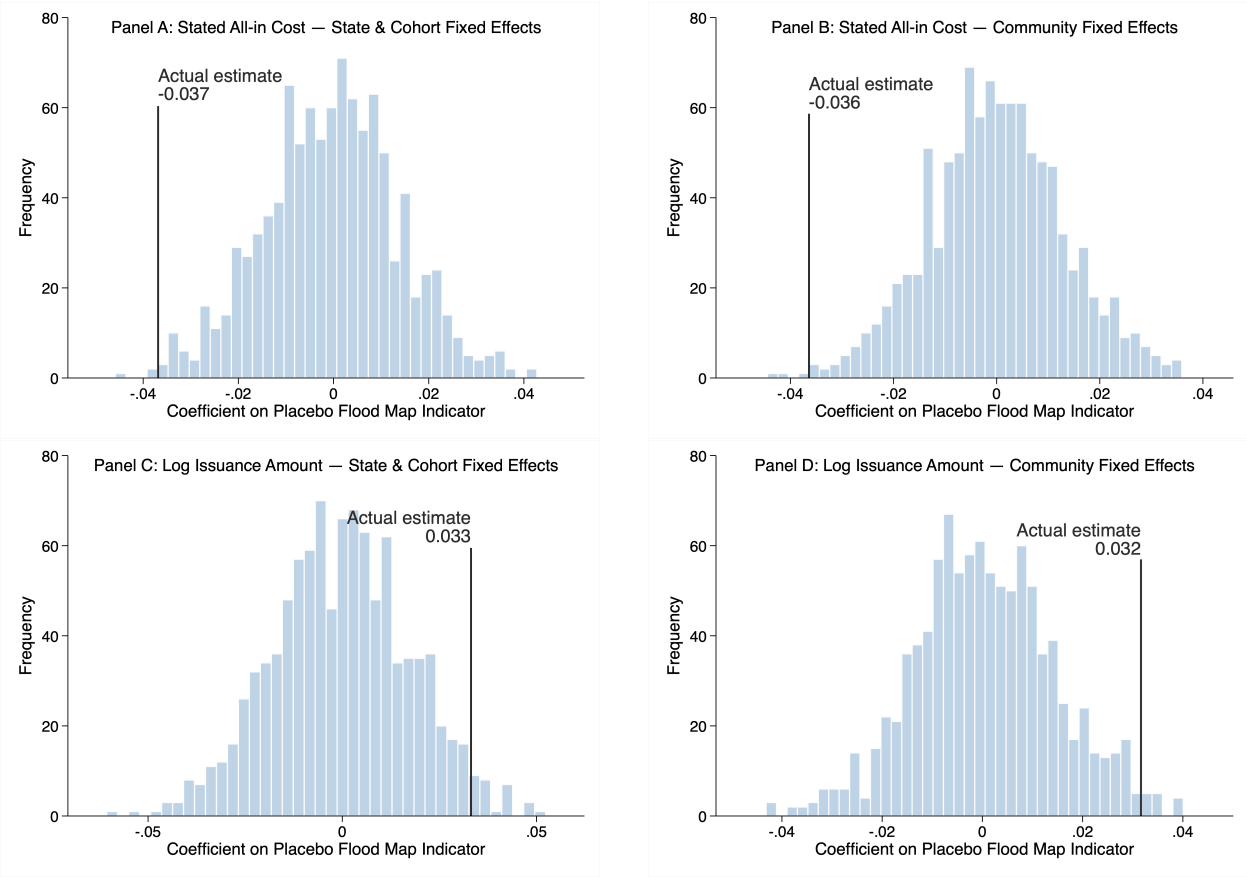


Figure C.5. Effect of Flood Insurance Access on Stated All-in Cost

This figure illustrates the effect of full flood insurance access on a municipality's stated all-in cost relative to other bond characteristics. Full flood insurance access is achieved when all communities within a county have an effective flood map, allowing residents to purchase federally subsidized flood insurance through the National Flood Insurance Program (NFIP).

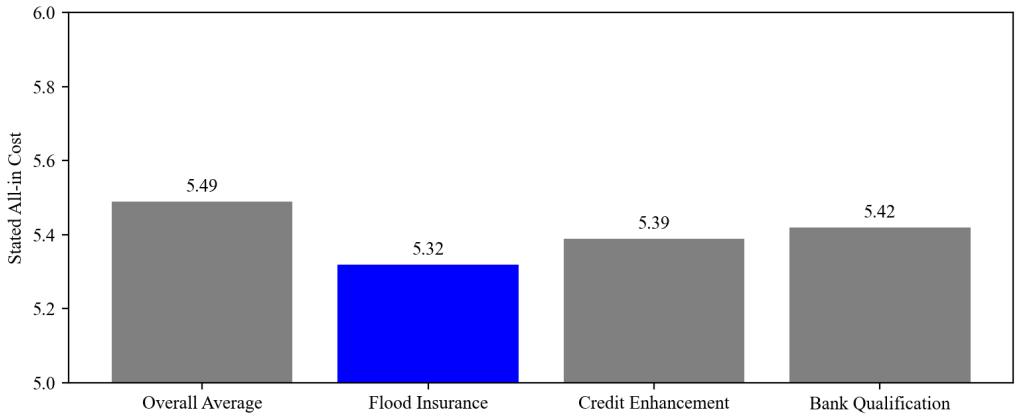


Table C.1. City-Matched Communities: Effect of Flood Insurance Access on Borrowing Costs and Amounts

This table presents OLS estimates of staggered difference-in-differences tests restricting the sample to National Flood Insurance Program (NFIP) communities matched to municipal bond issuers via exact or fuzzy city name matching, excluding county-fallback matches. The sample is collapsed to the community-by-month-year level for the stated all-in cost regressions and to the community-by-year level for the issuance amount regressions. The sample is further restricted to general obligation bonds with maturities greater than one year and a stated all-in cost below 50 percent, if populated. In Columns 1–2, the outcome variable is *Stated All-in Cost*$_{i,t}$, the average annual cost of debt the issuer pays on the bond, including underwriting and other fees. In Columns 3–4, the outcome variable is *Log Issuance Amount*$_{i,t}$, the log of the total issuance amount at the community level each year. *Treated*$_{i,t}$ is a binary variable equal to *Flood Insurance Access*$_{i,t}$, one if the community has received its initial flood map by the bond sale date, and zero otherwise. Bond and county control variables are the same as those in Table 3. Robust standard errors clustered at the community level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	Stated All-in Cost		Log Issuance Amount	
	(1)	(2)	(3)	(4)
Flood Insurance Access$_{i,t}$	-0.043***	-0.024*	0.048**	0.041**
	(0.015)	(0.014)	(0.021)	(0.017)
Observations	59,656	59,078	108,330	109,567
R-squared	0.914	0.933	0.478	0.732
Month-Year FE	Yes	Yes	No	No
Year FE	No	No	Yes	Yes
Bond & County Controls	Yes	Yes	Yes	Yes
State & Cohort FE	Yes	No	Yes	No
Community FE	No	Yes	No	Yes
Number of Communities	7,543	5,982	9,681	8,665
Mean of Dependent Variable	5.434	5.418	1.331	1.323

Table C.2. Dynamic Effect of Floods on Property Tax Revenue by Level of Flood Insurance Access

This table presents OLS estimates of staggered difference-in-differences tests that add flood exposure and its interaction with flood insurance access at horizons of zero through two years. The specifications are: $Y_{i,t} = \alpha + \sum_{k=0}^{2} \left(\beta_{1,k}\, Access_{i,t-k-1} \times F_{i,[t-k-1,\,t-k]} + \beta_{2,k}\, F_{i,[t-k-1,\,t-k]}\right) + \beta_3 \mathbf{X}_{i,t} + \delta_m + \delta_c + \delta_t + \varepsilon_{i,t}$ and $Y_{i,t} = \alpha + \sum_{k=0}^{2} \left(\beta_{1,k}\, Access_{i,t-k-1} \times F_{i,[t-k-1,\,t-k]} + \beta_{2,k}\, F_{i,[t-k-1,\,t-k]}\right) + \beta_3 \mathbf{X}_{i,t} + \delta_i + \delta_t + \varepsilon_{i,t}$. Subscript i represents the county, subscript m represents the state, subscript c represents the five-year FEMA mapping cohort, and subscript t represents the year. The observations are at the county-by-year level. The outcome variable is $Log\ Property\ Tax\ Revenue_{i,t}$, the log of annual county-level property tax revenue. $Access_{i,t-k-1}$ is $Flood\ Insurance\ Access$ measured at $12(k+1)$ months before the fiscal year end. In Columns 1–2, $F_{i,[t-k-1,\,t-k]}$ is $Flood_{[t-k-1,\,t-k]}$, a standardized count of the number of floods in the county during horizon k. In Columns 3–4, $F_{i,[t-k-1,\,t-k]}$ is $Log\ Flood\ Damage_{[t-k-1,\,t-k]}$, the log of total flood-related property damage in the county over the same horizon. Each specification also includes the base effect of $Flood\ Insurance\ Access_{i,t-1}$ through $Flood\ Insurance\ Access_{i,t-3}$ at each horizon; these terms are estimated but not tabulated for brevity. County control variables are the same as those in Table 2. Robust standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	Log Property Tax Revenue			
	(1)	(2)	(3)	(4)
$\text{Flood}_{[t-1,\,t]}$	-0.017***	-0.009***		
	(0.003)	(0.002)		
$\text{Flood}_{[t-1,\,t]} \times$ Insurance Access_{t-1}	0.032***	0.026***		
	(0.004)	(0.004)		
$\text{Flood}_{[t-2,\,t-1]}$	-0.014***	-0.004**		
	(0.003)	(0.002)		
$\text{Flood}_{[t-2,\,t-1]} \times$ Insurance Access_{t-2}	0.031***	0.019***		
	(0.004)	(0.003)		
$\text{Flood}_{[t-3,\,t-2]}$	-0.008***	-0.002		
	(0.003)	(0.002)		
$\text{Flood}_{[t-3,\,t-2]} \times$ Insurance Access_{t-3}	0.024***	0.019***		
	(0.005)	(0.004)		
Log Flood $\text{Damage}_{[t-1,\,t]}$			-0.003***	-0.002***
			(0.001)	(0.000)
Log Flood $\text{Damage}_{[t-1,\,t]} \times$ Insurance Access_{t-1}			0.006***	0.005***
			(0.001)	(0.001)
Log Flood $\text{Damage}_{[t-2,\,t-1]}$			-0.002***	-0.001**
			(0.000)	(0.000)
Log Flood $\text{Damage}_{[t-2,\,t-1]} \times$ Insurance Access_{t-2}			0.006***	0.004***
			(0.001)	(0.001)
Log Flood $\text{Damage}_{[t-3,\,t-2]}$			-0.001**	-0.000
			(0.001)	(0.000)
Log Flood $\text{Damage}_{[t-3,\,t-2]} \times$ Insurance Access_{t-3}			0.004***	0.003***
			(0.001)	(0.001)
Observations	101,460	107,600	101,460	107,600
R-squared	0.902	0.949	0.902	0.949
Year FE	Yes	Yes	Yes	Yes
County Controls	Yes	Yes	Yes	Yes
State & Cohort FE	Yes	No	Yes	No
County FE	No	Yes	No	Yes
Number of Counties	2,809	3,031	2,809	3,031
Mean of Dependent Variable	9.066	8.992	9.066	8.992

Table C.3. Effect of Flood Insurance Access on Stated All-in Cost by Flood Risk

This table presents OLS estimates of staggered difference-in-differences tests based on flood risk. The observations are at the community-by-month-year level. The sample is restricted to general obligation bonds with maturities greater than one year and a stated all-in cost below 50 percent, if populated. In Column 1, the sample is further restricted to exclude the communities that were mapped in the first six years (1969–1974) of the National Flood Insurance Program (NFIP). The outcome variable is *Stated All-in Cost*$_{i,t}$, the average annual cost of debt the issuer pays on the bond, including underwriting and other fees. *Flood Insurance Access*$_{i,t}$ is a binary variable equal to one if the community has received its initial flood map by the bond sale date and zero otherwise. *SFHA Fraction*$_i$ is the share of the community in a Special Flood Hazard Area. Bond and county control variables are the same as those in Table 3. Robust standard errors clustered at the community level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	Stated All-in Cost		
	(1)	(2)	(3)
Flood Insurance Access$_{i,t}$	-0.046***	-0.047***	-0.047***
	(0.016)	(0.015)	(0.016)
SFHA Fraction$_i$		0.058	
		(0.061)	
SFHA Fraction$_i$× Flood Insurance Access$_{i,t}$		0.088	0.100
		(0.069)	(0.072)
Observations	78,763	71,640	76,555
R-squared	0.915	0.914	0.933
Month-Year FE	Yes	Yes	Yes
State FE	Yes	Yes	No
Cohort FE	No	Yes	No
Community FE	No	No	Yes
Number of Communities	9,641	8,892	7,433
Mean of Dependent Variable	5.485	5.476	5.474

Table C.4. Effect of Placebo and Randomized Flood Insurance Access on Stated All-in Cost

This table presents OLS estimates of staggered difference-in-differences tests using randomized flood map dates. The observations are at the community-by-month-year level. The sample is restricted to general obligation bonds with maturities greater than one year and a stated all-in cost below 50 percent, if populated. In Column 1, *Placebo*: 1-3 $Years\ Before\ Mapping_{i,t}$ is an indicator for the three years immediately before the actual flood map date, controlling for the real $Flood\ Insurance\ Access_{i,t}$ indicator. Columns 2–3 replace each community's actual flood map date with a randomly drawn flood map date from the empirical distribution and re-estimate the main specification. The outcome variable is $Stated\ All\text{-}in\ Cost_{i,t}$, the average annual cost of debt the issuer pays on the bond, including underwriting and other fees. Bond and county control variables are the same as those in Table 3. Robust standard errors clustered at the community level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	Stated All-in Cost		
	(1)	(2)	(3)
Placebo: 1-3 Years Before Mapping$_{i,t}$	0.009		
	(0.013)		
Placebo: Random Flood Map Dates$_{i,t}$		-0.004	-0.004
		(0.012)	(0.016)
Flood Insurance Access$_{i,t}$	-0.033**		
	(0.017)		
Observations	76,989	76,989	76,555
R-squared	0.914	0.914	0.933
Month-Year FE	Yes	Yes	Yes
Bond & County Controls	Yes	Yes	Yes
State & Cohort FE	Yes	Yes	No
Community FE	No	No	Yes
Number of Communities	9,254	9,254	7,433
Mean of Dependent Variable	5.487	5.487	5.474

Table C.5. Event-Time: Effect of Flood Insurance Access on Stated All-in Costs

This table presents OLS estimates of event-time specifications around initial flood map issuance, following Deryugina (2017). The observations are at the community-by-month-year level. The sample is restricted to general obligation bonds with maturities greater than one year and a stated all-in cost below 50 percent, if populated. The sample is further restricted to observations within ten years before or after initial flood map issuance. In Columns 1–2, *Mapped 0–4 Years Ago*$_{i,t}$ and *Mapped 5–10 Years Ago*$_{i,t}$ are indicators for being mapped 0–4 years ago and 5–10 years ago. The omitted category is the pre-mapping portion of the event window. In Columns 3–4, $Mapped$ (0-10 $Years)_{i,t}$ is an indicator for being in the first ten years after initial flood map issuance. $Mapped \times Years\ Since_{i,t}$ interacts this post-mapping indicator with the number of years since initial mapping. The outcome variable is *Stated All-in Cost*$_{i,t}$, the average annual cost of debt the issuer pays on the bond, including underwriting and other fees. Control variables are the same as those in Table 3, augmented with climate controls that include monthly precipitation, precipitation squared, monthly mean temperature, and temperature squared at the county level to be consistent with Deryugina (2017). Robust standard errors clustered at the community level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	Stated All-in Cost			
	(1)	(2)	(3)	(4)
Mapped 0–4 Years Ago$_{i,t}$	-0.043*	-0.053**		
	(0.023)	(0.021)		
Mapped 5–10 Years Ago$_{i,t}$	-0.061*	-0.060*		
	(0.036)	(0.031)		
Mapped (0–10 Years)$_{i,t}$			-0.049**	-0.051**
			(0.021)	(0.021)
Mapped × Years Since$_{i,t}$			-0.004	0.000
			(0.004)	(0.004)
Observations	35,576	33,294	35,576	33,294
R-squared	0.854	0.888	0.854	0.888
Year FE	Yes	Yes	Yes	Yes
State & Cohort FE	Yes	No	Yes	No
Community FE	No	Yes	No	Yes
Climate Controls	Yes	Yes	Yes	Yes
Trend Control	No	No	Linear	Linear
Number of Communities	7,099	4,801	7,099	4,801
Mean of Dependent Variable	6.414	6.420	6.414	6.420

Table C.6. Effect of National Flood Insurance Program (NFIP) Claim Payouts on Stated All-in Cost

This table presents OLS estimates of staggered difference-in-differences tests for the interaction of floods with NFIP claim payouts. The observations are at the community-by-month-year level. The sample is restricted to general obligation bonds with maturities greater than one year and a stated all-in cost below 50 percent, if populated. The outcome variable is *Stated All-in Cost*$_{i,t}$, the average annual cost of debt the issuer pays on the bond, including underwriting and other fees. *Log Claim Payouts*$_{i,t-1}$ is the log of one plus CPI-adjusted NFIP claim payouts to the community over the prior 12 months, included both directly and interacted with *Flood*$_{i,t-1}$, the standardized count of floods in the community's county over the same period. Bond and county control variables are the same as those in Table 3. Robust standard errors clustered at the community level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	Stated All-in Cost	
	(1)	(2)
Flood$_{i,t-1}$	0.012***	0.009***
	(0.004)	(0.003)
Log Claim Payouts$_{i,t-1}$	0.003**	0.004***
	(0.001)	(0.001)
Flood$_{i,t-1}$× Log Claim Payouts$_{i,t-1}$	-0.001	-0.003***
	(0.001)	(0.001)
Observations	76,989	76,555
R-squared	0.914	0.933
Month-Year FE	Yes	Yes
Bond & County Controls	Yes	Yes
State & Cohort FE	Yes	No
Community FE	No	Yes
Number of Communities	9,254	7,433
Mean of Dependent Variable	5.487	5.474

Table C.7. Effect of Snowstorms on Stated All-in Cost

This table presents OLS estimates of staggered difference-in-differences tests for the interaction of flood insurance access with snowstorms, a second placebo hazard not covered by federal flood insurance. The specifications are: $Y_{i,t} = \alpha + \beta_1 \, Treated_{i,t-1} \times Snowstorm_{i,t-1} + \beta_2 \, Treated_{i,t-1} + \beta_3 \, Snowstorm_{i,t-1} + \beta_4 \mathbf{X}_{i,t} + \delta_m + \delta_c + \delta_t + \varepsilon_{i,t}$ and $Y_{i,t} = \alpha + \beta_1 \, Treated_{i,t-1} \times Snowstorm_{i,t-1} + \beta_2 \, Treated_{i,t-1} + \beta_3 \, Snowstorm_{i,t-1} + \beta_4 \mathbf{X}_{i,t} + \delta_i + \delta_t + \varepsilon_{i,t}$. Subscript i represents the community, subscript m represents the state, subscript c represents the five-year FEMA mapping cohort, and subscript t represents the month-year. The observations are at the community-by-month-year level. The sample is restricted to general obligation bonds with maturities greater than one year and a stated all-in cost below 50 percent, if populated. The outcome variable is *Stated All-in Cost*$_{i,t}$, the average annual cost of debt the issuer pays on the bond, including underwriting and other fees. $Treated_{i,t-1}$ is a binary variable representing *Flood Insurance Access*$_{i,t-1}$, one if the community had received its initial flood map 12 months before the sale date and zero otherwise. In Columns 1–2, *Snowstorm*$_{i,t-1}$ is the standardized count of snowstorms in the community's county in the prior 12 months. In Columns 3–4, *Log Snow Damage*$_{i,t-1}$ is the log of total snowstorm-related property damage in the community's county over the same period. Bond and county control variables are the same as those in Table 3. Robust standard errors clustered at the community level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	Stated All-in Cost			
	(1)	(2)	(3)	(4)
Flood Insurance Access$_{i,t-1}$	-0.044**	-0.040**	-0.047**	-0.043**
	(0.018)	(0.017)	(0.018)	(0.017)
Snowstorm$_{i,t-1}$	0.004	0.003		
	(0.004)	(0.004)		
Snowstorm$_{i,t-1}\times$ Flood Insurance Access$_{i,t-1}$	-0.003	-0.007		
	(0.007)	(0.006)		
Log Snow Damage$_{i,t-1}$			-0.000	-0.000
			(0.001)	(0.001)
Log Snow Damage$_{i,t-1}\times$ Access$_{i,t-1}$			0.000	-0.000
			(0.001)	(0.001)
Observations	76,989	76,555	76,989	76,555
R-squared	0.914	0.933	0.914	0.933
Month-Year FE	Yes	Yes	Yes	Yes
Bond & County Controls	Yes	Yes	Yes	Yes
State & Cohort FE	Yes	No	Yes	No
Community FE	No	Yes	No	Yes
Number of Communities	9,254	7,433	9,254	7,433
Mean of Dependent Variable	5.487	5.474	5.487	5.474